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# SCALABLE ITERATIVE DATA-ADAPTIVE RKHS REGULARIZATION*

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**Abstract.** We present iDARR, a scalable iterative data-adaptive RKHS regularization method for solving ill-posed linear inverse problems. This method searches for solutions in subspaces where the true solution can be identified, with the data-adaptive reproducing kernel Hilbert space (RKHS) penalizing the spaces of small singular values. At the core of the method is a new generalized Golub–Kahan bidiagonalization procedure that recursively constructs orthonormal bases for a sequence of RKHS-restricted Krylov subspaces. The method is scalable, with a complexity of $O(kmn)$ for $m$-by-$n$ matrices, where $k$ denotes the number of iterations. Numerical tests on the Fredholm integral equation and two-dimensional image deblurring demonstrate that it outperforms the widely used $L^2$ and $l^2$ norms, consistently producing stable and accurate solutions that converge when the noise level decreases.

**Key words.** iterative regularization, ill-posed inverse problem, reproducing kernel Hilbert space, Golub–Kahan bidiagonalization, deconvolution

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See reproducibility of computational results at end of the article.

## 1. Introduction.

This study considers large-scale ill-posed linear statistical inverse problems with limited prior information to determine an appropriate regularization norm. The task is to recover a high-dimensional solution $x \in \mathbb{R}^n$ from the equation

$$Ax + \mathbf{w} = b, \quad A \in \mathbb{R}^{m\times n}, \tag{1.1}$$

where $A$ and $b$ are a data-dependent forward map and noisy output, and $\mathbf{w}$ denotes noise or measurement error. In these problems, $A$ has singular values typically gradually decaying toward zero or even is rank deficient, rendering the least squares solution with minimal Euclidean norm, which is $x_{LS} = (A^\top A)^\dagger A^\top b$ with $^\dagger$ denoting

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the Moore–Penrose pseudoinverse of a matrix [13], highly sensitive to noise and small perturbations in the data. Such ill-posed problems arise in numerous scientific and engineering applications, ranging from reconstructing functions in functional linear regression [40, 44] to identifying integral kernels in nonlocal operators [22, 26].

Regularization is crucial for producing a robust solution for these ill-posed problems. In general, it comprises two components: (i) a penalty term that defines the admissible solution space, and (ii) a regularization parameter that balances data fidelity against that penalty. Broadly speaking, methods can be categorized into two main types: direct and iterative. Direct methods perform an explicit spectral decomposition of the matrix, with classical examples including Tikhonov regularization [37] and truncated SVD [11, 15]. Iterative methods, by contrast, construct approximate solutions via repeated matrix-vector multiplications, avoiding full decompositions and scaling more gracefully to large problems [7, 12, 45].

Since little is known a priori about the regularity of the true solution, we have two central challenges in regularization. (i) Adaptive norm selection: a fixed penalty (e.g., the Euclidean norm [15, 37] or the total variation norm [35]) cannot in general exploit the problem-specific geometry encoded by $A$. Although data-adaptive norms for nonparametric regression have been proposed [26, 27], they are defined implicitly and require an expensive spectral decomposition of $A^\top A$. (ii) Scalable implementation: even with a suitable norm, direct methods cost $O(n^3 + mn^2)$, which is prohibitive when $n$ and $m$ are large.

To address both challenges, we introduce iDARR, an iterative data-adaptive reproducing kernel Hilbert space regularization method. This method resolves both challenges by iteratively solving the subspace-projected problem

$$x_k = \operatorname*{argmin}_{x \in \mathcal{X}_k} \|x\|_{C_{rkhs}}, \quad \mathcal{X}_k = \left\{ x : x \in \operatorname*{argmin}_{\widetilde{x} \in \mathcal{S}_k} \|A\widetilde{x} - b\|_2 \right\},$$

where $\|\cdot\|_{C_{rkhs}}$ is the implicitly defined seminorm of a *data-adaptive reproducing kernel Hilbert space* (DA-RKHS), and $\mathcal{S}_k$ is a $k$-dimensional *solution subspace* within that RKHS. We construct $\mathcal{S}_k$ so that, as $k$ increases, it captures progressively more of the solution in the dominant directions of $A$ relative to the RKHS geometry through the seminorm $\|\cdot\|_{C_{rkhs}}$. Crucially, we impose an early stopping criterion to halt the iteration once additional dimensions begin to reflect more noise than signal, thereby preventing overfitting and ensuring that the iteration number $k$ serves as an effective regularization parameter. As a result, iDARR can produce a stable and accurate solution without using any matrix decompositions.

The DA-RKHS is a space defined adaptive to $A$. By reformulating $Ax + w = b$ as a weighted Fredholm integral equation of the first kind (see section 2), we show that the closure of the DA-RKHS coincides with the identifiable subspace of $A$. Thus, regularizing in this RKHS both confines the search to the right space and automatically penalizes small singular modes.

Our key innovation is a *generalized Golub–Kahan bidiagonalization* (gGKB) procedure, which iteratively constructs $\mathcal{S}_k$ for iDARR using only matrix-vector products. As a result, iDARR is scalable with a cost of only $O(kmn)$, where $k$ is the number of iterations, which is a dramatic savings compared to $O(n^3 + mn^2)$ for direct methods.

iDARR and gGKB have solid mathematical foundations. We prove that each subspace $\mathcal{S}_k$ is restricted in the DA-RKHS, thereby named the *RKHS-restricted Krylov subspaces,* and is spanned by the orthonormal vectors produced by gGKB. Importantly, if not stopped early, gGKB terminates when this subspace is fully explored, and iDARR recovers the unique least squares solution in it. Furthermore, we show that the gGKB method is a variant of the conjugate gradient (CG) method for

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linear inverse problems, thereby inheriting its well-known semiconvergence behavior and theoretical guarantees (see, e.g., [11, 15]).

Finally, we validate iDARR on prototypical Fredholm integral equations, with both exponential and polynomial spectral decay, as well as on large-scale two-dimensional (2D) image deblurring. In every case, iDARR outperforms state-of-the-art iterative regularizers employing $l^2$ and $L^2$ norms from the IR Tools package [12], producing reconstruction errors that decay reliably with the noise level and demonstrating robust, scalable performance.

**1.1. Related work.** Numerous regularization methods have been developed, and the literature on this topic is too vast to survey here; we refer to [11, 12, 15] and the references therein for an overview. In the following, we compare iDARR with the most closely related works.

*Regularization norms.* Various regularization norms are used for different applications, such as the Euclidean norm used in standard Tikhonov regularization [15, 37], the total variation norm $\|x'\|_{L^1}$ used in the Rudin–Osher–Fatemi method [35], the $L^1$ norm $\|x\|_{L^1}$ used in LASSO [36], and the RKHS norm $\|x\|_R^2$ of an RKHS with a user-specified reproducing kernel $R$ [3, 9, 39]. These norms, however, are often based on presumed properties of the solution and do not consider the specifics of each inverse problem. Our RKHS norm differs by adapting to the model and data: our RKHS has a reproducing kernel determined by the inverse problem, and its closure is the space in which the solution can be identified, making it an apt choice for regularization in the absence of additional solution information.

*Iterative regularization methods.* Iterative regularization (IR) methods are scalable by accessing the matrix only via matrix-vector multiplications, producing a sequence of estimators until early stopping, where the iteration number plays the role of the regularization parameter. IR has a rich history and remains a vibrant area of interest in contemporary studies [1, 12, 25, 32]. Different regularization terms lead to various methods. The LSQR algorithm [5, 32] with early stopping is standard for $\|x\|_2^2$-regularization. It solves projected problems in Krylov subspaces before transforming back to the original domain. For $\|Lx\|_2^2$ with $L \in \mathbb{R}^{p\times n}$, the widely used methods include the joint bidiagonalization method [18, 19], the generalized Krylov subspace method [21, 33], the random SVD or generalized SVD method [41, 42, 43], the modified truncated SVD method [2, 17], etc. For the general regularization norm of the form $x^\top Mx$ with a symmetric matrix $M$, the MLSQR in [1] treats positive definite $M$, and the preconditioned GKB in [25] handles positive semidefinite $M$. Our iDARR studies the case that $M$ is unavailable but $M^\dagger$ is ready to be used.

*Golub–Kahan bidiagonalization.* GKB was first used to solve inverse problems in [31], which generates orthonormal bases for Krylov subspaces in $(\mathbb{R}^n, \langle\cdot,\cdot\rangle_2)$ and $(\mathbb{R}^m, \langle\cdot,\cdot\rangle_2)$. This method extends to bounded linear compact operators between Hilbert spaces, with properties and convergence results detailed in [6]. Our gGKB extends the method to construct RKHS-restricted Krylov subspaces in $(\mathcal{R}(C_{rkhs}), \langle\cdot,\cdot\rangle_{C_{rkhs}})$, where $\mathcal{R}(C_{rkhs})$ is the range space of a positive semidefinite $C_{rkhs}$, and in particular, only $C_{rkhs}^\dagger$ is available.

The remainder of this paper is organized as follows. Section 2 introduces the adaptive RKHS with a characterization of its norm. Section 3 presents in detail the iDARR method. Section 4 presents several properties of iDARR to illustrate its regularization effect. In section 5, we systematically examine the algorithm and demonstrate the robust convergence of the estimator as the noise level decreases. Finally, section 6 concludes with a discussion on future developments.

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**2. A data-adaptive RKHS for regularization.** We now introduce an RKHS tailored to the matrix $A$. We refer to this space as a data-adaptive RKHS (DA-RKHS) because, in applications, $A$ is determined by the forward operator and data, as in functional linear regression [40, 44] and in learning integral kernels in operators [22, 26].

We connect the discrete system (1.1) with these applications by rewriting it as a weighted Fredholm integral equation of the first kind, so that $A^\top A$ corresponds to a normal operator, whose integral kernel $\overline{G}$ serves as the reproducing kernel of the DA-RKHS $H_G$. Importantly, minimizing the DA-RKHS norm $\|\cdot\|_{H_G}$ both confines the solution to the identifiable space and automatically penalizes the eigenspaces of small singular values.

After introducing the RKHS, we conclude this section with a brief review of the DARTR algorithm, a Tikhonov-style regularizer in the DA-RKHS introduced in [27], which motivates our scalable, iterative gGKB approach in section 3.

**2.1. Connecting the discrete system with a weighted Fredholm equation.** We recast the discrete system (1.1) as a weighted Fredholm integral equation of the first kind. Specifically, let $\mathcal{S} \subset \mathbb{R}$ be either a closed interval or a finite set, equipped with a finite measure $\nu$ (atomic when $\mathcal{S}$ is finite). Let $\mathcal{T} = \{t_j\}_{j=1}^m \subset \mathbb{R}$ with $0 = t_0 < t_1 < \cdots < t_m$ and equip $\mathcal{T}$ with the atomic measure $\mu$. Denote the associated square-integrable function spaces by $L^2_\nu(\mathcal{S})$ and $L^2_\mu(\mathcal{T})$, respectively. Let $K : \mathcal{S} \times \mathcal{T} \to \mathbb{R}$ be a continuous function (with $\mathcal{T}$ endowed with the discrete topology, and $\mathcal{S}$ when it is finite), and let $y(t)$ denote the noisy output of the weighted Fredholm equation,

$$y(t) = \int_{\mathcal{S}} K(t,s)\phi(s)\nu(ds) + \sigma w(t) \quad \forall t \in \mathcal{T}, \tag{2.1}$$

where $w(t)$ denotes measurement noise such that $\{w(t_j)\}$ are independent standard Gaussian random variables.

Consider the problem of estimating the function $\phi : \mathcal{S} \to \mathbb{R}$ in the form of $\phi = \sum_{i=1}^n x_i \phi_i$, where $\{\phi_i\}_{i=1}^n$ is a preselected collection of basis functions, from the noisy data $b = (y(t_1), \ldots, y(t_m))^\top \in \mathbb{R}^m$. Then, substituting this expansion into (2.1), we obtain (1.1) with $x = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$ and the matrix $A$ whose entries are

$$A(j,i) = \int_{\mathcal{S}} K(t_j, s)\phi_i(s)\nu(ds), \quad 1 \le j \le m, 1 \le i \le n. \tag{2.2}$$

In this study, we set the basis functions $\{\phi_i\}$ to be piecewise-constant and obtain $A$ as follows.

- When $\mathcal{S} = \{s_i\}_{i=1}^n$ with $\nu$ being an atomic measure, we set $\phi_i(s) = \mathbf{1}_{\{s_i\}}(s)$. Then, $\phi = x$ and the matrix $A$ has entries $A(j,i) = K(t_j, s_i)\nu(s_i)$.
- When $\mathcal{S}$ is a closed interval $[s_0, s_n]$ with $\nu$ being the Lebesgue measure, we set $\phi_i(s) = \mathbf{1}_{[s_{i-1}, s_i]}(s)$ with $s_0 < s_1 < \cdots < s_n$. Then, $\phi = \sum_{i=1}^n x_i \phi_i$ and the matrix $A$ has entries $A(j,i) = K(t_j, s_i)(s_i - s_{i-1})$.

Under the above settings, we rewrite the model (2.1) as

$$b = L_K \phi + \mathbf{w}, \tag{2.3}$$

where $L_K : L^2_\nu(\mathcal{S}) \to L^2_\mu(\mathcal{T})$ is the linear operator mapping $\phi$ to the discrete observation $b$, and $\mathbf{w}$ is the noise vector. A least squares solution is equivalent to a minimizer of the following loss function $\mathcal{E}(\phi)$ over $L^2_\nu(\mathcal{S})$:

$$\mathcal{E}(\phi) := \|L_K\phi - b\|^2_{L^2_\mu(\mathcal{T})} = \langle L_K\phi, L_K\phi\rangle_{L^2_\mu(\mathcal{T})} - 2\langle L_K\phi, b\rangle_{L^2_\mu(\mathcal{T})} + \|b\|^2_{L^2_\mu(\mathcal{T})}. \tag{2.4}$$

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The (weighted) Fredholm equation (2.1) is a prototype of linear ill-posed inverse problems, dating back to Hadamard [14], and it remains a test bed for new regularization methods [15, 24, 30, 38]. A common fix is to impose an $L^2_\nu(\mathcal{S})$ penalty, but this choice treats all directions equally and entirely neglects the null-space and small-eigenvalue structure of $L_K$. As a result, reconstructions can blow up in the vanishing-noise limit [23]. To overcome this instability, and in particular to design robust iterative regularizers, we introduce a weighted function space and an RKHS that are intrinsically adapted to both the data and the operator. These adaptive spaces, developed below, automatically confine the solution to the identifiable subspace of $L_K$ and penalize the ill-posed modes.

**2.2. The function space of identifiability.** We first introduce a weighted function space $L^2_\rho(\mathcal{S})$, where the measure $\rho$ is defined as

$$\frac{d\rho}{d\nu}(s) := \frac{1}{Z}\int_{\mathcal{T}} |K(t,s)|\mu(dt)\,\forall s \in \mathcal{S}, \tag{2.5}$$

where $Z = \int_{\mathcal{S}}\int_{\mathcal{T}} |K(t,s)|\mu(dt)\nu(ds)$ is a normalizing constant. For the discrete model (1.1), the exploration measure is the normalized column sum of the absolute values of the matrix $A$. This measure quantifies the exploration of data to the unknown function $\phi$ through the integral kernel $K$ at the output set $\mathcal{T}$, i.e., $\{K(t_j,\cdot)\}_{t_j\in\mathcal{T}}$, hence it is referred to as an *exploration measure.* For example, if $\int_{\mathcal{T}} |K(t,s)|\mu(dt) = 0$ for some $s \in \mathcal{S}$ (e.g., when the column $s$ of $A$ is zero), then $\frac{d\rho}{d\nu}(s) = 0$, indicating that the data provides no information about $\phi(s)$.

We measure the estimation error of $\phi$ in the weighted space $L^2_\rho(\mathcal{S})$ rather than $L^2_\nu(\mathcal{S})$ because $L^2_\rho(\mathcal{S})$ induces a problem-specific and data-dependent metric. In particular, if $\frac{d\rho}{d\nu}(s) = 0$ for some $s \in \mathcal{S}$ (so the data contain no information about $\phi(s)$), the error at $s$ is ignored in $L^2_\rho(\mathcal{S})$ but still contributes in $L^2_\nu(\mathcal{S})$, making the latter unsuitable for evaluating regularized estimators. For notation simplicity, hereafter we write $L^2_\rho(\mathcal{S})$ and $L^2_{\rho\otimes\rho}(\mathcal{S}\times\mathcal{S})$ as $L^2_\rho$ and $L^2_{\rho\otimes\rho}$, respectively.

Next, we consider the variational inverse problem over $L^2_\rho$, and the goal is to find a minimizer of the loss function (2.4) in $L^2_\rho$. Since the loss function is quadratic, its Hessian is a symmetric positive linear operator, and it has a unique minimizer in the linear subspace where the Hessian is strictly positive. We assign a name to this subspace in the next definition.

DEFINITION 2.1 (function space of identifiability). *In a variational inverse problem of minimizing a quadratic loss function $\mathcal{E}$ in $L^2_\rho$, we call $\mathcal{L}_{\overline{G}} = \frac{1}{2}\nabla^2\mathcal{E}$ the* normal operator*, where $\nabla^2\mathcal{E}$ is the Hessian of $\mathcal{E}$, and we call $H = \mathrm{Null}(\mathcal{L}_{\overline{G}})^\perp$ the* function space of identifiability (FSOI)*.*

The next lemma specifies the FSOI for the loss function in (2.4). We use $P_H$ to denote the orthogonal projector onto a closed subspace $H$.

LEMMA 2.2. *Assume $K \in C(\mathcal{T}\times\mathcal{S})$. For $\rho$ in* (2.5)*, define $\overline{G}: \mathcal{S}\times\mathcal{S} \to \mathbb{R}$ as*

$$\overline{G}(s,s') := \frac{G(s,s')}{\frac{d\rho}{d\nu}(s)\frac{d\rho}{d\nu}(s')}, \quad G(s,s') := \int_{\mathcal{T}} K(t,s)K(t,s')\mu(dt). \tag{2.6}$$

(a) *The function $\overline{G} \in L^2_{\rho\otimes\rho}$ is a symmetric positive definite kernel. The normal operator for $\mathcal{E}$ in* (2.4) *over $L^2_\rho$ is $\mathcal{L}_{\overline{G}}: L^2_\rho \to L^2_\rho$ defined by*

$$\mathcal{L}_{\overline{G}}\phi(s) := \int_{\mathcal{S}} \phi(s')\overline{G}(s,s')\rho(ds'), \tag{2.7}$$

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*and the loss function can be written as*

$$\mathcal{E}(\phi) = \langle \mathcal{L}_{\overline{G}}\phi, \phi\rangle_{L^2_\rho} - 2\left\langle \phi^D, \phi\right\rangle_{L^2_\rho} + \text{const.}, \tag{2.8}$$

*where $\phi^D$ comes from the Riesz representation theorem, s.t. $\left\langle \phi^D, \phi\right\rangle_{L^2_\rho} = \langle L_K\phi, b\rangle_{L^2_\mu(\mathcal{T})}$ for any $\phi \in L^2_\rho$.*

(b) *$\mathcal{L}_{\overline{G}}$ is compact, self-adjoint, and positive. Hence, its eigenvalues $\{\lambda_i\}_{i\geq 1}$ converge to zero and its orthonormal eigenfunctions $\{\psi_i\}_i$ form a complete basis of $L^2_\rho(\mathcal{S})$.*

(c) *The FSOI is $H := \overline{\text{span}\{\psi_i\}_{i:\lambda_i>0}} \subset L^2_\rho(\mathcal{S})$. When the data is noiseless, i.e., $b = L_K\phi_{\text{true}}$, the unique minimizer of $\mathcal{E}$ in $H$ is $\widehat{\phi} = {\mathcal{L}_{\overline{G}}}^{-1}\phi^D = P_H\phi_{\text{true}}$, where ${\mathcal{L}_{\overline{G}}}^{-1}$ is defined on $\mathcal{L}_{\overline{G}}(L^2_\rho)$.*

*Proof.* (a) Since $K$ is continuous, and $\mathcal{T}$ and $\mathcal{S}$ are compact, we can assume $|K(s,t)| \leq C_K$ on $\mathcal{T}\times\mathcal{S}$ with a positive scalar $C_K$. It is clear that $G$ and $\overline{G}$ are symmetric. First, we show that $\overline{G}$ is square-integrable. Since

$$G(s,s') \leq C_K \int_{\mathcal{T}} |K(t,s)|\mu(dt) \leq ZC_K\dot{\rho}(s),$$

by the symmetry of $G$, we obtain that $G(s,s') \leq ZC_K\min\{\dot{\rho}(s), \dot{\rho}(s')\}$ for any $s, s' \in \mathcal{S}$, where $\dot{\rho}(s) := \frac{d\rho}{d\nu}(s)$ for notational simplicity. Now we have

$$\int_{\mathcal{S}}\int_{\mathcal{S}} \overline{G}(s,s')^2\rho(ds)\rho(ds') = \int_{\mathcal{S}}\int_{\mathcal{S}} \frac{G(s,s')^2}{\dot{\rho}(s)\dot{\rho}(s')}\nu(ds)\nu(ds') \leq (ZC_K)^2\nu(\mathcal{S})^2 < \infty,$$

where $\nu(\mathcal{S})$ is the measure of $\mathcal{S}$. For any $\{c_j\}_{j=1}^n \subset \mathbb{R}$, $\{s_j\}_{j=1}^k \subset \mathcal{S}$, and $n \in \mathbb{N}$, since

$$\sum_{i=1}^n\sum_{j=1}^n c_ic_j\overline{G}(s_i,s_j) = \int_{\mathcal{T}}\left(\sum_{i=1}^n c_i\mathbf{1}_{\mathcal{S}}(s_i)K(t,s_i)/\dot{\rho}(s_i)\right)^2\mu(dt) \geq 0,$$

then $\overline{G}$ is positive definite. Thus, $\mathcal{L}_{\overline{G}}$ is a compact self-adjoint operator.

For any $\phi, \psi \in L^2_\rho$, it holds that

$$\begin{aligned}
\langle L_K\phi, L_K\psi\rangle_{L^2_\mu(\mathcal{T})} &= \int_{\mathcal{T}}\left(\int_{\mathcal{S}} K(t,s)\phi(s)\nu(ds)\int_{\mathcal{S}} K(t,s')\psi(s')\nu(ds')\right)\mu(dt)\\
&= \int_{\mathcal{S}}\left(\int_{\mathcal{S}}\left(\int_{\mathcal{T}} K(t,s)K(t,s')\mu(dt)\right)\phi(s)\nu(ds)\right)\psi(s')\nu(ds')\\
&= \int_{\mathcal{S}}\left(\int_{\mathcal{S}} \overline{G}(s,s')\phi(s)\rho(ds)\right)\psi(s')\rho(ds')\\
&= \langle \mathcal{L}_{\overline{G}}\phi, \psi\rangle_{L^2_\rho}.
\end{aligned} \tag{2.9}$$

This implies that the normal operator of $\mathcal{E}$ is $\mathcal{L}_{\overline{G}}$. For (2.8), it remains to prove the second term in the right-hand side. Since $\phi \mapsto \langle L_K\phi, b\rangle_{L^2_\mu(\mathcal{T})}$ is a bounded linear functional on $L^2_\rho$, the Riesz representation theorem gives $\left\langle \phi^D, \phi\right\rangle_{L^2_\rho} = \langle L_K\phi, b\rangle_{L^2_\mu(\mathcal{T})}$ with $\phi^D \in L^2_\rho$.

(b) Since $\mathcal{L}_{\overline{G}}$ is compact and self-adjoint, the claim follows immediately from the spectral theory for compact self-adjoint operators.

(c) Since the normal operator of $\mathcal{E}$ is $\mathcal{L}_{\overline{G}}$, the FSOI is $\text{Null}(\mathcal{L}_{\overline{G}})^\perp = \overline{\text{span}\{\psi_i\}_{i:\lambda_i>0}}$ by the spectral theory, where the closure is taken in $L^2_\rho$. The Fréchet derivative $\nabla\mathcal{E}(\phi)$ in $L^2_\rho$ satisfies

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$$\langle \nabla \mathcal{E}(\phi), \psi \rangle_{L^2_\rho(\mathcal{S})} = \lim_{h \to 0} \frac{\mathcal{E}(\phi + h\psi) - \mathcal{E}(\phi)}{h} = \left\langle 2(\mathcal{L}_{\overline{G}}\phi - \phi^D), \psi \right\rangle_{L^2_\rho(\mathcal{S})} \forall \psi \in L^2_\rho(\mathcal{S}).$$

Thus, $\nabla \mathcal{E}(\phi) = 2(\mathcal{L}_{\overline{G}}\phi - \phi^D)$. When the data is noiseless, we have

$$\left\langle \phi^D, \phi \right\rangle_{L^2_\rho} = \langle L_K \phi_{\text{true}}, L_K \phi \rangle_{L^2_\mu(\mathcal{T})} = \left\langle \mathcal{L}_{\overline{G}} \phi_{\text{true}}, \phi \right\rangle_{L^2_\rho}.$$

Thus, $\phi^D = \mathcal{L}_{\overline{G}}\phi_{\text{true}}$. Since $\mathcal{L}_{\overline{G}}(P_{H^\perp}\phi_{\text{true}}) = \mathcal{L}_{\overline{G}}(P_{\text{Null}(\mathcal{L}_{\overline{G}})}\phi_{\text{true}}) = 0$, we have $\nabla \mathcal{E}(P_H \phi_{\text{true}}) = 2(\mathcal{L}_{\overline{G}}\phi_{\text{true}} - \phi^D) = 0$. This implies that $\widehat{\phi} = P_H \phi_{\text{true}}$. ∎

Lemma 2.2 reveals the cause of the ill-posedness and provides insights on regularization:

- The variational inverse problem has a unique minimizer only in the FSOI $H$, which can be a proper subset of $L^2_\rho$. Its ill-posedness in $H$ depends on the smallest eigenvalue of the operator $\mathcal{L}_{\overline{G}}$ and the error in $\phi^D$.
- When the data is noiseless, one can only identify the $H$-projection of the true input function. When data is noisy, its minimizer $\mathcal{L}_{\overline{G}}^{-1}\phi^D$ is ill-defined in $L^2_\rho$ when $\phi^D \notin \mathcal{L}_{\overline{G}}(L^2_\rho)$, which may happen when $H$ is infinite-dimensional.

As a result, when regularizing the ill-posed problem, a crucial task is to ensure that the search takes place in the FSOI and to filter out the noise that contaminates the components in the FSOI $H$.

**2.3. A data-adaptive RKHS.** Our *data-adaptive RKHS* is defined by the reproducing kernel $\overline{G}$ in (2.6), so that it automatically incorporates both the model's integral kernel $K$ and the observed data. Regularizing with its norm confines the estimator to the RKHS, which is a subspace of the FSOI, while simultaneously suppressing directions associated with small eigenvalues. Concretely, if $\phi = \sum_{k \geq 1} c_k \psi_k$, where $\{\psi_k\}$ and $\{\lambda_k\}$ are $\mathcal{L}_{\overline{G}}$'s orthonormal eigenfunctions and eigenvalues, then $\|\phi\|^2_{H_G} = \sum_{k \geq 1} \lambda_k^{-1} c_k^2$ (see the next lemma), so components with small $\lambda_k$ incur a larger penalty.

The next lemma characterizes the properties of this RKHS.

LEMMA 2.3 (characterization of the DA-RKHS). *Assume $K \in C(\mathcal{T} \times \mathcal{S})$. The RKHS $H_G$ with $\overline{G}$ as its reproducing kernel satisfies the following properties.*

(a) *$H_G := \mathcal{L}_{\overline{G}}^{\frac{1}{2}}(L^2_\rho(\mathcal{S}))$ has inner product $\langle \phi, \phi \rangle_{H_G} = \langle \mathcal{L}_{\overline{G}}^{-\frac{1}{2}}\phi, \mathcal{L}_{\overline{G}}^{-\frac{1}{2}}\phi \rangle_{L^2_\rho(\mathcal{S})}$.*

(b) *$\{\sqrt{\lambda_i}\psi_i\}_{\lambda_i > 0}$ is an orthonormal basis of $H_G$, where $\{(\lambda_i, \psi_i)\}_i$ are eigenpairs of $\mathcal{L}_{\overline{G}}$.*

(c) *For any $\phi = \sum_{i \geq 1} c_i \psi_i \in H_G$, we have*

$$\langle L_K \phi, L_K \phi \rangle_{L^2_\mu(\mathcal{T})} = \sum_{i=1}^{\infty} \lambda_i c_i^2, \quad \|\phi\|^2_{L^2_\rho} = \sum_{i=1}^{\infty} c_i^2, \quad \|\phi\|^2_{H_G} = \sum_{i=1}^{\infty} \lambda_i^{-1} c_i^2.$$

(d) *$H = \overline{H_G}$ with closure in $L^2_\rho(\mathcal{S})$, where $H = \overline{\text{span}\{\psi_i\}_{i:\lambda_i > 0}}$ is the FSOI.*

*Proof.* Parts (a) and (b) are the standard characterization theorem of RKHS; see, e.g., [9, section 4.4] or [27].

Part (c) follows from the identity (2.9) and the claim of (b).

Part (d) follows from that $\overline{H_G} = \overline{\text{span}\{\sqrt{\lambda_i}\psi_i\}_{i:\lambda_i > 0}} = \overline{\text{span}\{\psi_i\}_{i:\lambda_i > 0}} = H$. ∎

The next theorem shows the computation of the RKHS norm for the problem (1.1) when it is written in the form (2.1)–(2.2). A key component is solving the eigenvalues of $\mathcal{L}_{\overline{G}}$ through a generalized eigenvalue problem.

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THEOREM 2.4 (computation of the RKHS norm). *Suppose that* (1.1) *is equivalent to* (2.1) *under basis functions* $\{\phi_i\}_{i=1}^n$ *with* $n \le \infty$ *and* (2.2). *Let* $B$ *with entries* $B(i,j) = \langle \phi_i, \phi_j \rangle_{L^2_\rho}$ *be a nonsingular basis matrix, where* $\rho$ *is the measure defined in* (2.5). *Then, the operator* $\mathcal{L}_{\overline{G}}$ *in* (2.7) *has eigenvalues* $(\lambda_1, \dots, \lambda_n)$ *solved by the generalized eigenvalue problem:*

$$A^\top A V = B V \Lambda \quad \text{s.t.} \quad V^\top B V = I_n, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \tag{2.10}$$

*with eigenfunctions* $\{\psi_k = \sum_{j=1}^n V_{jk}\phi_j\}_{k=1}^n$. *The RKHS norm of* $\phi = \sum_{i=1}^n x_i \phi_i$ *satisfies*

$$\begin{aligned} &\|\phi\|^2_{H_G} = \|x\|^2_{C_{rkhs}} = x^\top C_{rkhs} x, \\ &C^\dagger_{rkhs} = V \Lambda V^\top = B^{-1}(A^\top A) B^{-1}. \end{aligned} \tag{2.11}$$

*In particular, if* $B = I_n$, *we have* $C^\dagger_{rkhs} = A^\top A$.

*Proof.* Denote $\mathbf{\Phi} = (\phi_1, \dots, \phi_n)^\top$ and $\mathbf{\Psi} = (\psi_1, \dots, \psi_n)^\top$. We first prove that the eigenvalues of $\mathcal{L}_{\overline{G}}$ are solved by (2.10). We suppose $\{(\lambda_i, \psi_i)\}_{i=1}^n$ are the eigenvalues and eigenfunctions of $\mathcal{L}_{\overline{G}}$ over $L^2_\rho$ with $\{\psi_i\}$ being an orthonormal basis of $L^2_\rho$. Since $\operatorname{span}\{\phi_i\}_{i=1}^n \supseteq \mathcal{L}_{\overline{G}}(L^2_\rho)$, there exists $V \in \mathbb{R}^{n\times n}$ such that $\mathbf{\Psi} = V^\top \mathbf{\Phi}$, i.e., $\psi_k = \sum_{j=1}^n V_{jk}\phi_j$ for each $1 \le k \le n$. Then, the task is to verify that $V$ and $\Lambda = \operatorname{Diag}(\lambda_1, \dots, \lambda_n)$ satisfy $A^\top A V = B V \Lambda$ and $V^\top B V = I_n$.

The orthonormality of $\{\psi_i\}$ implies that

$$I_n = \left( \langle \psi_k, \psi_l \rangle_{L^2_\rho} \right)_{1\le k,l\le n} = \left( \left\langle \sum_{i=1}^n V_{ik}\phi_i, \sum_{j=1}^n V_{jl}\phi_j \right\rangle_{L^2_\rho} \right)_{1\le k,l\le n} = V^\top B V.$$

Note that $[A^\top A](j,i) = \langle L_K \phi_i, L_K \phi_j \rangle_{L^2_{\mu}(\mathcal{T})} = \langle \phi_j, \mathcal{L}_{\overline{G}} \phi_i \rangle_{L^2_\rho}$ for $1 \le i, j \le n$. Then,

$$\langle \phi_j, \mathcal{L}_{\overline{G}} \psi_k \rangle_{L^2_\rho} = \sum_{i=1}^n [A^\top A](j,i) V_{ik}.$$

Meanwhile, the eigenequation $\mathcal{L}_{\overline{G}} \psi_k = \lambda_k \psi_k$ implies that for each $\phi_j$,

$$\langle \phi_j, \mathcal{L}_{\overline{G}} \psi_k \rangle_{L^2_\rho} = \lambda_k \langle \phi_j, \psi_k \rangle_{L^2_\rho} = \lambda_k \left\langle \phi_j, \sum_{i=1}^n V_{ik}\phi_i \right\rangle_{L^2_\rho} = \lambda_k \sum_{i=1}^n B_{ji} V_{ik},$$

i.e., $(\langle \phi_j, \mathcal{L}_{\overline{G}} \psi_k \rangle_{L^2_\rho}) = B V \Lambda$. Hence, these two equations imply that $A^\top A V = B V \Lambda$.

Next, to compute the norm of $\phi = \sum_{i=1}^n x_i \phi_i \in H_G$, we write it as $\phi = x^\top \mathbf{\Phi} = x^\top V^{-1} \mathbf{\Psi}$. Then, its norm is

$$\|\phi\|^2_{H_G} = \sum_{k=1}^n \lambda_k^{-1} \mathbf{1}_{\lambda_k > 0} \left( x^\top V^{-1} \right)_k^2 = x^\top V^{-1} \Lambda^\dagger V^{-\top} x = x^\top (V \Lambda V^\top)^\dagger x.$$

Thus, we have $C_{rkhs} = (V \Lambda V^\top)^\dagger$. Let $\widetilde{V} = B^{\frac{1}{2}} V$. Then (2.10) implies that $\widetilde{V}^\top \widetilde{V} = I_n$ and

$$(B^{-\frac{1}{2}} A^\top A B^{-\frac{1}{2}}) \widetilde{V} = \widetilde{V} \Lambda.$$

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It then follows that

$$V\Lambda V^\top = B^{-\frac{1}{2}}\widetilde{V}\Lambda\widetilde{V}^\top B^{-\frac{1}{2}} = B^{-\frac{1}{2}}(B^{-\frac{1}{2}}A^\top AB^{-\frac{1}{2}})B^{-\frac{1}{2}} = B^{-1}(A^\top A)B^{-1},$$

and consequently, $C_{rkhs}^\dagger = V\Lambda V^\top = B^{-1}(A^\top A)B^{-1}$. ∎

In particular, when (1.1) is either a discrete model or a discretization of (2.1) based on Riemann sum approximation of the integral, the exploration measure $\rho$ is the normalized column sum of the absolute values of the matrix $A$, and $B = \mathrm{diag}(\rho)$. See subsection 5.1 for details.

**2.4. DARTR: Data-adaptive RKHS Tikhonov regularization.** We review DARTR, a data-adaptive RKHS Tikhonov regularization algorithm introduced in [27]. Specifically, it solves problem (1.1) with DA-RKHS regularization:

$$x_\lambda = \underset{x\in\mathcal{R}(C_{rkhs})}{\mathrm{argmin}} \ \mathcal{E}_\lambda(x) := \|Ax-b\|_2^2 + \lambda\|x\|_{C_{rksh}}^2,$$

where the norm $\|\cdot\|_{C_{rksh}}$ is the DA-RKHS norm introduced in Theorem 2.4, and the regularization parameter $\lambda > 0$ is selected to balance the loss $\|Ax_\lambda - b\|_2^2$ and the estimator norm $\|x_\lambda\|_{C_{rksh}}^2$. The minimizer $x_\lambda$ is solved from $(A^\top A + \lambda C_{rkhs})x = A^\top b$ in $\mathcal{R}(C_{rkhs})$.

DARTR introduces a transformation matrix $C_* := V\Lambda^{1/2}$ to avoid using the numerically unstable pseudoinverse in $C_{rkhs}$. Note that $C_*^\top C_{rkhs} C_* = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) =: \mathbf{I}_r$, where $I_r$ is the identity matrix with rank $r$, the number of positive eigenvalues in $\Lambda$. Then, the linear equation $(A^\top A + \lambda C_{rkhs})x = A^\top b$ with $x \in \mathcal{R}(C_{rkhs})$ is equivalent to

$$(C_* A^\top A C_* + \lambda \mathbf{I}_r)\widetilde{x} = C_* A^\top b \tag{2.12}$$

with $x = C_*\widetilde{x}$. Thus, DARTR computes $\widetilde{x}_\lambda$ in the above equation by least squares with minimal norm and returns $x_\lambda = C_*\widetilde{x}_\lambda$ with $\lambda$ selected by the L-curve method [16].

DARTR is a direct method based on matrix decompositions, and it takes $O(mn^2 + n^3)$ flops. Hence, it is computationally infeasible when $n$ is large. The iterative method in the next section implements the RKHS regularization in a scalable fashion.

**3. Iterative regularization by DA-RKHS.** This section introduces a subspace project method tailored to utilize the DA-RKHS for IR. As an iterative method, it achieves scalability by accessing the coefficient matrix only via matrix-vector multiplications, producing a sequence of estimators until reaching a desired solution. This section follows the notation conventions in Table 1.

**3.1. Overview.** Our regularization method is based on subspace projection in the DA-RKHS. It iteratively constructs a sequence of linear subspaces $\mathcal{S}_k$ of the DA-RKHS $(\mathcal{R}(C_{rkhs}), \langle\cdot,\cdot\rangle_{C_{rkhs}})$ and recursively solves the projected problems

$$x_k = \underset{x\in\mathcal{X}_k}{\mathrm{argmin}} \|x\|_{C_{rkhs}}, \quad \mathcal{X}_k = \left\{ x : x \in \underset{\widetilde{x}\in\mathcal{S}_k\in\mathcal{R}(C_{rkhs})}{\mathrm{argmin}} \|A\widetilde{x} - b\|_2 \right\}. \tag{3.1}$$

TABLE 1
*Table of notation.*

| | |
|---|---|
| $A, B, C$ | matrix or array by capital letters |
| $b, c, x, y, z, u, v$ | vector by lowercase letters |
| $\alpha, \beta, \gamma$ | scalar by Greek letters |
| $\mathcal{R}(A)$ and $\mathcal{N}(A)$ | the range and null spaces of matrix $A$ |

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The subspaces $\mathcal{S}_k$, which we call the *solution subspaces,* are designed to grow incrementally, gradually capturing more of the dominant features encoded by the RKHS seminorm $\|\cdot\|_{C_{rkhs}}$. The iterations will be stopped early to avoid $\mathcal{S}_k$ from accumulating too many noisy components. We will propose an approach to iteratively construct $\mathcal{S}_k$. This approach ensures the uniqueness of the solution within each iteration, as detailed in Theorem 4.6, thereby yielding a sequence of solutions $\{x_k\}$ with each emerging from its corresponding subspace. The iterations proceed until it meets an early stopping criterion, designed to exclude excessively noisy components and thereby achieve effective regularization. In this framework, the iteration number $k$ serves as the regularization parameter.

This method is a recursion of the following three parts:

P1. *Construct the solution subspaces.* We introduce a new gGKB to construct the solution subspaces in the DA-RKHS iteratively. The procedure is presented in Algorithm 3.1.

P2. *Recursively update the solution to the projected problem.* We solve the least squares problem in the solution subspaces in (3.1) efficiently by a new LSQR-type algorithm, Algorithm 3.2, updating $\|x_k\|_{C_{rkhs}}$ and the residual norm $\|Ax-b\|_2$ without even computing the residual.

P3. *Regularize by early stopping.* We select the optimal $k$ by either the discrepancy principle (DP) when we have an accurate estimate of $\|\mathbf{w}\|_2$ or the L-curve criterion otherwise.

In the next subsection, we present details for these key parts. Then, we analyze the computational complexity of the algorithm.

**3.2. Algorithm details and derivations.** This subsection presents detailed derivations for the three parts P1–P3 in Algorithm 3.3.

**P1. Construct the solution subspaces.** We construct the solution subspaces by incorporating the regularization term $\|\cdot\|^2_{C_{rkhs}}$ in the GKB process. A key point is to use $C^{\dagger}_{rkhs} = B^{-1}A^{\top}AB^{-1}$ to avoid explicitly computing $C_{rkhs}$, which involves the costly spectral decomposition of the normal operator; see (2.11) in Theorem 2.4.

Consider first the case where $C_{rkhs}$ is positive definite. In this scenario, $A$ has full column rank, and the $C_{rkhs}$-inner product Hilbert space $(\mathbb{R}^n, \langle\cdot,\cdot\rangle_{C_{rkhs}})$ is a discrete representation of the RKHS $H_G$ with the given basis functions. Note that the true solution is mapped to the noisy $b$ by $A : (\mathbb{R}^n, \langle\cdot,\cdot\rangle_{C_{rkhs}}) \to (\mathbb{R}^m, \langle\cdot,\cdot\rangle_2)$. Let $A^* : (\mathbb{R}^m, \langle\cdot,\cdot\rangle_2) \to (\mathbb{R}^n, \langle\cdot,\cdot\rangle_{C_{rkhs}})$ be the adjoint of $A$, i.e., $\langle Ax, b\rangle_2 = \langle x, A^*b\rangle_{C_{rkhs}}$ for any $x \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$. By definition, the matrix-form expression of $A^*$ is

$$A^* = C^{-1}_{rkhs}A^{\top}, \tag{3.2}$$

since $\langle Ax, b\rangle_2 = x^{\top}A^{\top}b$ and $\langle x, A^*b\rangle_{C_{rkhs}} = x^{\top}C_{rkhs}A^*b$ for any $x$ and $b$.

The GKB process recursively constructs orthonormal bases for these two Hilbert spaces starting with the vector $b$ as follows:

$$\beta_1 u_1 = b, \quad \alpha_1 z_1 = A^* u_1, \tag{3.3a}$$

$$\beta_{i+1} u_{i+1} = Az_i - \alpha_i u_i, \tag{3.3b}$$

$$\alpha_{i+1} z_{i+1} = A^* u_{i+1} - \beta_{i+1} z_i, \tag{3.3c}$$

where $u_i \in (\mathbb{R}^m, \langle\cdot,\cdot\rangle_2)$, $z_i \in (\mathbb{R}^n, \langle\cdot,\cdot\rangle_{C_{rkhs}})$ with $z_0 = \mathbf{0}$, and $\alpha_i$, $\beta_i$ are normalizing factors such that $\|u_i\|_2 = \|z_i\|_{C_{rkhs}} = 1$. The iteration starts with $u_1 = b/\beta_1$ with $\beta_1 = \|b\|_2$. Using $A^*$ in (3.2), we write (3.3c) as

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$$\alpha_{i+1}z_{i+1} = C_{rkhs}^{-1}A^\top u_{i+1} - \beta_{i+1}z_i \tag{3.4}$$

with $\alpha_{i+1} = \|C_{rkhs}^{-1}A^\top u_{i+1} - \beta_{i+1}z_i\|_{C_{rkhs}}$. To compute $\alpha_{i+1}$ without explicitly computing $C_{rkhs}$, define $\bar{z}_i = C_{rkhs}z_i$. Then we have

$$\alpha_{i+1}\bar{z}_{i+1} = A^\top u_{i+1} - \beta_{i+1}\bar{z}_i, \tag{3.5}$$

where $\bar{z}_0 := \mathbf{0}$. Let $p = A^\top u_{i+1} - \beta_{i+1}\bar{z}_i$. Then we obtain $\alpha_{i+1} = \|C_{rkhs}^{-1}p\|_{C_{rkhs}} = (p^\top C_{rkhs}^{-1}p)^{1/2}$, which uses $C_{rkhs}^{-1} = B^{-1}A^\top AB^{-1}$ without computing $C_{rkhs}$.

Next, consider that $C_{rkhs}$ is positive semidefinite. The iterative process remains the same with $C_{rkhs}^{-1}$ replaced by the pseudoinverse $C_{rkhs}^\dagger$, because $C_{rkhs}^\dagger = B^{-1}A^\top AB^{-1}$ has the same form as $C_{rkhs}^{-1}$ for the nonsingular case. Specifically, the recursive relation (3.4) becomes

$$\alpha_{i+1}z_{i+1} = C_{rkhs}^\dagger A^\top u_{i+1} - \beta_{i+1}z_i. \tag{3.6}$$

To compute $\alpha_{i+1}$, we use the property that $z_i \in \mathcal{R}(C_{rkhs})$, which will be proved in Proposition 4.3. Note that $C_{rkhs}^\dagger C_{rkhs} = P_{\mathcal{N}(C_{rkhs})^\perp} = P_{\mathcal{R}(C_{rkhs})}$ since $C_{rkhs}$ is symmetric, where $P_{\mathcal{X}}$ is the projection operator onto subspace $\mathcal{X}$. It follows that $C_{rkhs}^\dagger C_{rkhs}z_i = z_i$. Therefore, (3.6) becomes

$$\alpha_{i+1}z_{i+1} = C_{rkhs}^\dagger(A^\top u_{i+1} - \beta_{i+1}C_{rkhs}z_i).$$

For $i = 1$, define $\bar{z}_i = A^\top u_1/\alpha$, which leads to $z_1 = C_{rkhs}^\dagger \bar{z}_1$. Suppose we already have computed $\bar{z}_i$ and $z_i = C_{rkhs}^\dagger \bar{z}_i$. It follows that $C_{rkhs}^\dagger C_{rkhs}z_i = C_{rkhs}^\dagger C_{rkhs}C_{rkhs}^\dagger \bar{z}_i = C_{rkhs}^\dagger \bar{z}_i$. Therefore, the above recursive relation can be rewritten as

$$\alpha_{i+1}z_{i+1} = C_{rkhs}^\dagger(A^\top u_{i+1} - \beta_{i+1}\bar{z}_i).$$

By letting $p = A^\top u_{i+1} - \beta_{i+1}\bar{z}_i$ and $\bar{z}_{i+1} = p/\alpha_{i+1}$, we get $z_{i+1} = C_{rkhs}^\dagger \bar{z}_{i+1}$ and $\alpha_{i+1} = \|C_{rkhs}^\dagger p\|_{C_{rkhs}} = (p^\top C_{rkhs}^\dagger p)^{1/2}$.

Thus, the two cases of $C_{rkhs}$ lead to the same iterative process. We summarize the iterative process in Algorithm 3.1 and call it *generalized Golub–Kahan bidiagonalization* (gGKB).

Suppose the gGKB process terminates at step $k_t := \max_{k\geq 1}\{\alpha_k\beta_k > 0\}$. We show in Propositions 4.4 and 4.5 that the output vectors $\{u_i\}_{i=1}^{k_t}$ and $\{z_i\}_{i=1}^{k_t}$ are orthonormal in $(\mathbb{R}^m, \langle\cdot,\cdot\rangle_2)$ and $(\mathbb{R}^n, \langle\cdot,\cdot\rangle_{C_{rkhs}})$, respectively. In particular, they

**Algorithm 3.1.** Generalized Golub–Kahan bidiagonalization (gGKB).

**Input:** $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $B = \mathrm{diag}(\rho)$

1: Initialize: let $\beta_1 = \|b\|_2$, $u_1 = b/\beta_1$, and compute $p = A^\top u_1$, $s = B^{-1}A^\top AB^{-1}p$.
2: Let $\alpha_1 = (s^\top p)^{1/2}$, $z_1 = s/\alpha_1$, $\bar{z}_1 = p/\alpha_1$.
3: **for** $i = 1, 2, \ldots, k$, **do**
4: $\quad r = Az_i - \alpha_i u_i$, $\beta_{i+1} = \|r\|_2$, $u_{i+1} = r/\beta_{i+1}$;
5: $\quad p = A^\top u_{i+1} - \beta_{i+1}\bar{z}_i$, $s = B^{-1}A^\top AB^{-1}p$; ▷ $C_{rkhs}^\dagger = B^{-1}A^\top AB^{-1}$
6: $\quad \alpha_{i+1} = (s^\top p)^{1/2}$, $z_{i+1} = s/\alpha_{i+1}$, $\bar{z}_{i+1} = p/\alpha_{i+1}$. ▷ $z_i = C_{rkhs}^\dagger \bar{z}_i$
7: **end for**

**Output:** $\{\alpha_i, \beta_i\}_{i=1}^{k+1}$, $\{u_i, z_i, \bar{z}_i\}_{i=1}^{k+1}$

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span two Krylov subspaces generated by $\{AC_{rkhs}^{\dagger}A^{\top}, b\}$ and $\{C_{rkhs}^{\dagger}A^{\top}A, C_{rkhs}^{\dagger}A^{\top}b\}$, respectively.

In matrix form, the $k$-step gGKB process with $k < k_t$, starting with vector $b$, produces a 2-orthonormal matrix $U_{k+1} = (u_1, \ldots, u_{k+1}) \in \mathbb{R}^{m\times(k+1)}$ (i.e., $U_{k+1}^{\top}U_{k+1} = I_{k+1}$), a $C_{rkhs}$-orthonormal matrix $Z_{k+1} = (z_1, \ldots, z_{k+1}) \in \mathbb{R}^{n\times(k+1)}$, and a bidiagonal matrix

$$B_k = \begin{pmatrix} \alpha_1 & & & \\ \beta_2 & \alpha_2 & & \\ & \beta_3 & \ddots & \\ & & \ddots & \alpha_k \\ & & & \beta_{k+1} \end{pmatrix} \in \mathbb{R}^{(k+1)\times k} \tag{3.7}$$

such that the recursion in (3.3a), (3.3b) and (3.6) can be written as

$$\beta_1 U_{k+1}e_1 = b, \tag{3.8a}$$

$$AZ_k = U_{k+1}B_k, \tag{3.8b}$$

$$C_{rkhs}^{\dagger}A^{\top}U_{k+1} = Z_kB_k^{\top} + \alpha_{k+1}z_{k+1}e_{k+1}^{\top}, \tag{3.8c}$$

where $e_1$ and $e_{k+1}$ are the first and $(k+1)$th columns of $I_{k+1}$. We emphasize that $B_k$ is a bidiagonal matrix of full column rank, since $\alpha_i, \beta_i > 0$ for all $i \le k+1$.

**P2. Recursively update the solution to the projected problem.** For each $k \le k_t$, i.e., before gGKB terminates, we solve (3.1) in the subspace

$$\mathcal{S}_k := \operatorname{span}\{z_1, \ldots, z_k\} \tag{3.9}$$

and compute the RKHS norm of the solution.

We show first the uniqueness of the solution to (3.1). From (3.8b) we have $B_k = U_{k+1}^{\top}AZ_k$, which implies that $B_k$ is a projection of $A$ onto the two subspaces $\operatorname{span}\{U_{k+1}\}$ and $\operatorname{span}\{Z_k\}$. Since any vector $x$ in $\mathcal{S}_k$ can be written as $x = Z_ky$ with a $y \in \mathbb{R}^k$, we obtain from (3.8a) and (3.8b) that for any $x = Z_ky$,

$$\begin{aligned} \min_{x=Z_ky} \|Ax - b\|_2 &= \min_{y\in\mathbb{R}^k} \|AZ_ky - U_{k+1}\beta_1e_1\|_2 \\ &= \min_{y\in\mathbb{R}^k} \|U_{k+1}B_ky - U_{k+1}\beta_1e_1\|_2 = \min_{y\in\mathbb{R}^k} \|B_ky - \beta_1e_1\|_2. \end{aligned} \tag{3.10}$$

Since $k \le k_t$, $B_k$ has full column rank, this $k$-dimensional least squares problem has a unique solution. Therefore, the unique solution to (3.1) is

$$x_k = Z_ky_k, \quad y_k = \operatorname*{argmin}_{y\in\mathbb{R}^k} \|B_ky - \beta_1e_1\|_2. \tag{3.11}$$

We note that the above uniqueness is independent of the specific construction of the basis vectors $\{z_k\}$. In general, as long as $\mathcal{S}_k \subseteq \mathcal{R}(C_{rkhs})$, the solution to (3.1) is unique; see Theorem 4.6.

Importantly, we recursively compute $x_k$ without explicitly solving $y_k$ in (3.11), avoiding the $O(k^3)$ computational cost. To this end, we adopt a procedure similar to the LSQR algorithm in [32] to iteratively update $x_k$ from $x_0 = \mathbf{0}$. It starts from the following Givens QR factorization:

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$$Q_k \begin{pmatrix} B_k & \beta_1 e_1 \end{pmatrix} = \begin{pmatrix} R_k & f_k \\ & \bar{\gamma}_{k+1} \end{pmatrix} := \left(\begin{array}{ccccc:c} \rho_1 & \theta_2 & & & & \gamma_1 \\ & \rho_2 & \theta_3 & & & \gamma_2 \\ & & \ddots & \ddots & & \vdots \\ & & & \rho_{k-1} & \theta_k & \gamma_{k-1} \\ & & & & \rho_k & \gamma_k \\ \hdashline & & & & & \bar{\gamma}_{k+1} \end{array}\right),$$

where the orthogonal matrix $Q_k$ is the product of a series of Givens rotation matrices, and $R_k$ is a bidiagonal upper triangular matrix; see [13, section 5.2.5]. We implement the Givens QR factorization using the procedure in [32], which recursively zeros out the subdiagonal elements $\beta_i$ for each $2 \le i \le k+1$. Specifically, at the $i$th step, a Givens rotation zeros out $\beta_{i+1}$ by

$$\begin{pmatrix} c_i & s_i \\ s_i & -c_i \end{pmatrix} \begin{pmatrix} \bar{\rho}_i & 0 & \bar{\gamma}_i \\ \beta_{i+1} & \alpha_{i+1} & 0 \end{pmatrix} = \begin{pmatrix} \rho_i & \theta_{i+1} & \gamma_i \\ 0 & \bar{\rho}_{i+1} & \bar{\gamma}_{i+1} \end{pmatrix},$$

where the entries $c_i$ and $s_i$ of the Givens rotations satisfy $c_i^2 + s_i^2 = 1$, and the elements $\rho_i$, $\theta_{i+1}$, $\bar{\rho}_{i+1}$, $\gamma_i$, $\bar{\gamma}_{i+1}$ are recursively updated accordingly; see [32] for more details.

As a result of the QR factorization, we can write

$$\|B_k y - \beta_1 e_1\|_2^2 = \left\| Q_k \begin{pmatrix} B_k & \beta_1 e_1 \end{pmatrix} \begin{pmatrix} y \\ -1 \end{pmatrix} \right\|_2^2 = \|R_k y - f_k\|_2^2 + |\bar{\gamma}_{k+1}|^2. \tag{3.12}$$

Hence, the solution to $\min_{y \in \mathbb{R}^k} \|B_k y - \beta_1 e_1\|$ is $y_k = R_k^{-1} f_k$. Factorizing $R_k$ as

$$R_k = D_k \bar{R}_k, \quad D_k := \begin{pmatrix} \rho_1 & & & \\ & \rho_2 & & \\ & & \ddots & \\ & & & \rho_k \end{pmatrix}, \ \bar{R}_k := \begin{pmatrix} 1 & \theta_2/\rho_1 & & \\ & 1 & \theta_3/\rho_2 & \\ & & \ddots & \theta_k/\rho_{k-1} \\ & & & 1 \end{pmatrix}$$

and denoting $W_k = Z_k \bar{R}_k^{-1} = (w_1, \ldots, w_k)$, we get

$$x_k = Z_k y_k = Z_k R_k^{-1} f_k = (Z_k \bar{R}_k^{-1})(D_k^{-1} f_k) = W_k (D_k^{-1} f_k) = \sum_{i=1}^{k} (\gamma_i/\rho_i) w_i.$$

Updating $x_k$ recursively, and solving $W_k \bar{R}_k = Z_k$ by back substitution, we obtain

$$x_i = x_{i-1} + (\gamma_i/\rho_i) w_i, \quad w_{i+1} = z_{i+1} - (\theta_{i+1}/\rho_i) w_i \quad \forall 1 \le i \le k. \tag{3.13}$$

Last, we compute $\|x_i\|_{C_{rkhs}}$ without an explicit $C_{rkhs}$. By using the property $z_i = C_{rkhs}^{\dagger} \bar{z}_i$ and recursively defining

$$\bar{x}_i = \bar{x}_{i-1} + (\gamma_i/\rho_i) \bar{w}_i, \quad \bar{w}_{i+1} = \bar{z}_{i+1} - (\theta_{i+1}/\rho_i) \bar{w}_i \quad \forall 1 \le i \le k,$$

where $\bar{x}_0 = \mathbf{0}$ and $\bar{w}_1 = \bar{z}_1$, we have $x_i = C_{rkhs}^{\dagger} \bar{x}_i$ and $\|x_i\|_{C_{rkhs}}^2 = x_i^{\top} \bar{x}_i$.

The updating procedure is described in Algorithm 3.2. This algorithm yields the residual norm $\|Ax_k - b\|_2$ without explicitly computing the residual. In fact, by (3.10) and (3.12) we have

$$\bar{\gamma}_{k+1} = \|B_k y_k - \beta_1 e_1\|_2 = \|Ax_k - b\|_2. \tag{3.14}$$

Note that $\bar{\gamma}_{k+1}$ decreases monotonically since $x_k$ minimizes $\|Ax - b\|_2$ in the gradually expanding subspace $\mathrm{span}\{Z_k\}$.

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**Algorithm 3.2.** Updating procedure.

Let $x_0 = \mathbf{0},\ \bar{x}_0 = \mathbf{0},\ w_1 = z_1,\ \bar{w}_1 = \bar{z}_1,\ \bar{\gamma}_1 = \beta_1,\ \bar{\rho}_1 = \alpha_1$
**for** $i = 1, 2, \ldots, k$, **do**
  $\rho_i = (\bar{\rho}_i^2 + \beta_{i+1}^2)^{1/2},\ c_i = \bar{\rho}_i/\rho_i,\ s_i = \beta_{i+1}/\rho_i$
  $\theta_{i+1} = s_i\alpha_{i+1},\ \bar{\rho}_{i+1} = -c_i\alpha_{i+1},\ \gamma_i = c_i\bar{\gamma}_i,\ \bar{\gamma}_{i+1} = s_i\bar{\gamma}_i$
  $x_i = x_{i-1} + (\gamma_i/\rho_i)w_i,\ w_{i+1} = z_{i+1} - (\theta_{i+1}/\rho_i)w_i$
  $\bar{x}_i = \bar{x}_{i-1} + (\gamma_i/\rho_i)\bar{w}_i,\ \bar{w}_{i+1} = \bar{z}_{i+1} - (\theta_{i+1}/\rho_i)\bar{w}_i$
  $\|x_i\|_{C_{rkhs}} = (x_i^\top \bar{x}_i)^{1/2}$
**end for**

Importantly, Algorithm 3.2 efficiently computes the solution $x_k$ and $\|x_k\|_{C_{rkhs}}$. At each step of updating $x_i$ or $w_{i+1}$, the computation takes $O(2n)$ flops. Similarly, for updating $\bar{x}_i$ or $\bar{w}_{i+1}$, as well as for computing $\|x_i\|_{C_{rkhs}}$, the number of flops is also $O(2n)$. Therefore, the dominant computational cost is $O(10n)$. In contrast, if $y_k$ is solved explicitly at each step, it takes $O(\sum_{i=1}^k i^3) \sim O(k^4)$ flops; together with the step of forming $x_k = Z_k y_k$ that takes $O(kn)$ flops, they lead to a total cost of $O(kn + k^4)$ flops. Thus, the LSQR-type iteration in Algorithm 3.2 significantly reduces the number of flops from $O(kn + k^4)$ to $O(10n)$.

**P3. Regularize by early stopping.** An early stopping strategy is imperative to prevent the solution subspace from becoming excessively large to contain too many noisy components, which could otherwise compromise the regularization. This necessity is rooted in the phenomenon of *semiconvergence*: the iteration vector $x_k$ initially approaches an optimal regularized solution but subsequently moves toward the unstable naive solution to $\min_{x\in\mathcal{R}(C_{rkhs})} \|Ax - b\|_2$, as demonstrated in Theorems 4.7 and 4.8.

For early stopping, we adopt the L-curve criterion, as outlined in [12]. This method identifies the ideal early stopping iteration $k_*$ at the corner of the curve represented by

$$(\log\|Ax_k - b\|_2, \|x_k\|_{C_{rkhs}}) = (\log\bar{\gamma}_{k+1}, \log\|x_k\|_{C_{rkhs}})\,. \tag{3.15}$$

Here $\bar{\gamma}_{k+1}$ and $\|x_k\|_{C_{rkhs}}$ are computed with negligible cost in Algorithm 3.2. To construct the L-curve effectively, we set gGKB to execute at least 10 iterations. Additionally, to enhance numerical stability, we terminate gGKB when either $\alpha_i$ or $\beta_i$ is close to machine precision, following Theorem 4.7.

It is noteworthy that the DP presents a viable alternative when the measurement error $\|\mathbf{w}\|_2$ in (1.1) is available with a high degree of accuracy. The DP halts iterations at the earliest instance of $k$ that satisfies $\bar{\gamma}_{k+1} = \|Ax_k - b\|_2 \le \tau\|\mathbf{w}\|_2$, where $\tau$ is chosen to be marginally greater than 1.

Overall, the iDARR procedure is presented in Algorithm 3.3.

In iDARR, the regularized solution is sensitive to the number of iterations, and the DP or L-curve criterion may yield a suboptimal number of iterations, leading to an over- or underregularized solution. To overcome the semiconvergence, we propose the hybrid iDARR (hyb-iDARR) method, which applies a $C_{rkhs}$ Tikhonov regularization to the projected problem at each iteration. For more details about the hybrid regularization method, we refer readers to [8, 20, 34]. At the $k$th iteration, with $\mathcal{S}_k = \text{span}\{Z_k\}$, hyb-iDARR solves the following projected problem:

$$\min_{x=Z_k y,\, y\in\mathbb{R}^k} \left\{\|Ax - b\|_2^2 + \lambda_k\|Z_k y\|_{C_{rkhs}}^2\right\} = \min_{y\in\mathbb{R}^k} \left\{\|B_k y - \beta_1 e_1\|_2^2 + \lambda_k\|y\|_2^2\right\},$$

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**Algorithm 3.3.** iDARR: Iterative data-adaptive RKHS regularization.

**Input:** $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $B = \mathrm{diag}(\rho)$, $x_0 = \mathbf{0}$, $\bar{x}_0 = \mathbf{0}$, where $\rho$ is the exploration measure in (2.5).

1: **for** $k = 1, 2, \ldots,$ **do**
2: P1. Compute $u_k$, $z_k$, $\bar{z}_k$, $\alpha_k$, $\beta_k$ by gGKB in Algorithm 3.1.
3: P2. Update $x_k$, $\bar{\gamma}_{k+1}$, $\|x_k\|_{C_{rkhs}}$, etc., by Algorithm 3.2.
4: P3: Stop at iteration $k_*$ if *Early stopping criterion* is satisfied. ▷ L-curve or DP
5: **end for**

**Output:** Final solution $x_{k_*}$

where $\lambda_k$ is the regularization parameter that can be updated iteratively using the weighted generalized cross-validation method [8, 25]. Therefore, at the $k$th iteration, the hyb-iDARR method yields the solution

$$x_{k,\lambda_k} = Z_k y_{k,\lambda_k}, \quad y_{k,\lambda_k} = \underset{y\in\mathbb{R}^k}{\mathrm{argmin}} \left\{ \|B_k y - \beta_1 e_1\|_2^2 + \lambda_k \|y\|_2^2 \right\}. \tag{3.16}$$

Note that the above regularization problem is only $k$-dimensional, making the computation highly efficient.

**3.3. Computational complexity.** Suppose the algorithm takes $k$ iterations and the basis matrix $B$ is diagonal. Recall that $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times n}$, $u_i \in \mathbb{R}^m$, and $z_i \in \mathbb{R}^n$. The total computational cost of Algorithm 3.3 is about $O(3mnk)$ when $m \le n/3$ or $k < n/3$ and about $O((m+k)n^2)$ when otherwise. The cost is dominated by the gGKB process since the cost of the update procedure in Algorithm 3.2 is only $O(n)$ at each step.

gGKB can be computed in two approaches. The first approach uses only matrix-vector multiplication. The main computations at each iteration occur at the matrix-vector products $p = A^\top u_i$ and $s = B^{-1}A^\top AB^{-1}p = B^{-1}(A^\top(A(B^{-1}p)))$, which take $O(mn)$ and $O(2mn)$ flops, respectively. Thus, the total computational cost of gGKB is $O(3mnk)$ flops. Another approach is using $\mathbf{A} = A^\top A$ instead of $A^\top$ and $A$ to compute $s$. In this approach, the computation of $\mathbf{A}$ from $A$ takes $O(mn^2)$ flops, and the matrix-vector multiplication $\mathbf{A}v$ in each iteration takes about $O(n^2)$ flops. Hence, the total cost of $k$ iterations is $O(mn^2 + kn^2)$. The second approach is faster when $mn^2 + n^2k < 3mnk$, or equivalently, $(3m - n)k > mn$. That is, roughly speaking, $m > n/3$ and $k > mn/(3m - n) > n/3$.

In practice, the matrix-vector computation is preferred since the iteration number $k$ is often small. The resulting iDARR algorithm requires approximately $O(3mnk)$ flops.

**4. Properties of iDARR.** This section studies the properties of the iDARR method, including the structure of the solution subspaces generated by gGKB, the orthogonality of the resulting vectors $\{u_i\}_{i=1}^{k+1}$ and $\{z_i\}_{i=1}^{k+1}$, and the number of iterations at termination, defined as

$$\textbf{gGKB terminate step:} \quad k_t := \max_{k\ge 1}\{\alpha_k\beta_k > 0\}. \tag{4.1}$$

Additionally, we present several properties of the iterative solutions, including the uniqueness of the solution at each step, a characterization of the solution at the terminate step, and its relationship to the CG method.

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Throughout this section, let $r$ denote the rank of $\Lambda$ and let $V_r$ denote the first $r$ columns of $V$, where $\Lambda$ and $V$ are matrices constituted by the generalized eigenvalues and eigenvectors of $\{A, B\}$ in (2.11). We have $\text{rank}(A) = r$. Recall that $C_{rkhs}^{\dagger} = B^{-1}(A^{\top}A)B^{-1} = V\Lambda V^{\top}$ is symmetric, leading to $\mathcal{R}(C_{rkhs}) = \mathcal{R}(C_{rkhs}^{\dagger}) = \mathcal{R}(V_r)$. Note that DA-RKHS is $(\mathcal{R}(C_{rkhs}), \langle\cdot,\cdot\rangle_{C_{rkhs}})$.

**4.1. Properties of the solution subspaces.** We show first that the gGKB-produced vectors $\{u_i\}_{i=1}^{k+1}$ and $\{z_i\}_{i=1}^{k+1}$ are orthogonal in $\mathbb{R}^n$ and in DA-RKHS, and the solution subspaces of gGKB are RKHS-restricted Krylov subspaces.

DEFINITION 4.1 (RKHS-restricted Krylov subspace). *Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^n$, and $B \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix. Let the discrete DA-RKHS be $(\mathcal{R}(C_{rkhs}), \langle\cdot,\cdot\rangle_{C_{rkhs}})$ with $C_{rkhs}^{\dagger} = B^{-1}(A^{\top}A)B^{-1}$. The RKHS-restricted Krylov subspaces are*

$$\mathcal{K}_{k+1}(C_{rkhs}^{\dagger}A^{\top}A, C_{rkhs}^{\dagger}A^{\top}b) = \text{span}\{(C_{rkhs}^{\dagger}A^{\top}A)^i C_{rkhs}^{\dagger}A^{\top}b\}_{i=0}^{k}, \quad k \geq 0. \tag{4.2}$$

The main result is the following.

THEOREM 4.2 (properties of gGKB). *Recall $k_t$ in (4.1), and gGKB generates vectors $\{u_i\}_{i=1}^{k}$, $\{z_i\}_{i=1}^{k}$ and $\mathcal{S}_k = \text{span}\{z_i\}_{i=1}^{k}$. They satisfy the following properties:*

(i) *$\{u_i\}_{i=1}^{k}$ and $\{z_i\}_{i=1}^{k}$ are orthonormal in $\mathbb{R}^n$ and in $(\mathcal{R}(C_{rkhs}), \langle\cdot,\cdot\rangle_{C_{rkhs}})$, respectively;*

(ii) *$\mathcal{S}_k = \mathcal{K}_k(C_{rkhs}^{\dagger}A^{\top}A, C_{rkhs}^{\dagger}A^{\top}b)$ for each $k \leq k_t$, and the termination iteration number is $k_t = \dim(\mathcal{K}_{\infty}(C_{rkhs}^{\dagger}A^{\top}A, C_{rkhs}^{\dagger}A^{\top}b))$.*

*Proof.* Part (i) follows from Proposition 4.3, where we show that $z_k \in \mathcal{R}(C_{rkhs})$, and Proposition 4.4, where we show the orthogonality of these vectors.

For part (ii), $\mathcal{S}_k = \mathcal{K}_k(C_{rkhs}^{\dagger}A^{\top}A, C_{rkhs}^{\dagger}A^{\top}b)$ follows from that $\{z_i\}_{i=1}^{k}$ form an orthonormal basis of the RKHS-restricted Krylov subspace. We prove $k_t = \dim(\mathcal{K}_{\infty}(C_{rkhs}^{\dagger}A^{\top}A, C_{rkhs}^{\dagger}A^{\top}b))$ in Proposition 4.5. ∎

PROPOSITION 4.3. *For each $z_i$ generated by gGKB in (3.3), it holds that $z_i \in \mathcal{R}(C_{rkhs})$. Additionally, if $q := C_{rkhs}^{\dagger}A^{\top}u_{i+1} - \beta_{i+1}z_i \neq \mathbf{0}$, then $\alpha_{i+1} = \|q\|_{C_{rkhs}} \neq 0$.*

*Proof.* We prove it by mathematical induction. For $i = 1$, we obtain from (3.6) and Theorem 2.4 that $\alpha_1 z_1 = C_{rkhs}^{\dagger}A^{\top}u_1 = V\Lambda V^{\top}A^{\top}u_1 \in \mathcal{R}(V_r) = \mathcal{R}(C_{rkhs})$, where $r$ is the rank of $\Lambda$. Suppose $z_i \in \mathcal{R}(C_{rkhs})$ for $i \geq 1$. Using again (3.6) and Theorem 2.4 we get

$$\alpha_{i+1}z_{i+1} = C_{rkhs}^{\dagger}A^{\top}u_{i+1} - \beta_{i+1}z_i = V\Lambda V^{\top}A^{\top}u_{i+1} - \beta_{i+1}z_i \in \mathcal{R}(C_{rkhs}).$$

Therefore, $z_{i+1} \in \mathcal{R}(C_{rkhs})$, and $q \in \mathcal{R}(C_{rkhs})$.

If $\alpha_{i+1} = 0$, then $q \in \mathcal{N}(C_{rkhs}) = \mathcal{R}(C_{rkhs})^{\perp}$. Therefore, $q = \mathbf{0}$. ∎

Thus, even if $C_{rkhs}$ is singular (positive semidefinite), the gGKB in Algorithm 3.1 does not terminate as long as the right-hand sides of (3.3b) and (3.6) are nonzero, since the iterative computation of $\{\beta_{i+1}, u_{i+1}\}$ and $\{\alpha_{i+1}, z_{i+1}\}$ can continue. Next, we show that these vectors are orthogonal.

PROPOSITION 4.4 (orthogonality). *Suppose the $k$-step gGKB does not terminate, i.e., $k < k_t$; then $\{u_i\}_{i=1}^{k+1}$ is a 2-orthonormal basis of the Krylov subspace*

$$\mathcal{K}_{k+1}(AC_{rkhs}^{\dagger}A^{\top}, b) = \text{span}\{(AC_{rkhs}^{\dagger}A^{\top})^i b\}_{i=0}^{k}, \tag{4.3}$$

*and $\{z_i\}_{i=1}^{k+1}$ is a $C_{rkhs}$-orthonormal basis of the RKHS-restricted Krylov subspace defined in (4.2).*

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*Proof.* Note from Theorem 2.4 that $C^{\dagger}_{rkhs} = V\Lambda V^{\top}$. Let $W_r = V_r\Lambda_r^{1/2}$. Then $W_r^{\top}C_{rkhs}W_r = I_r$, $C^{\dagger}_{rkhs} = W_rW_r^{\top}$, and $\mathcal{R}(W_r) = \mathcal{R}(C_{rkhs})$. For any $z_i$, Proposition 4.3 implies that there exists $v_i \in \mathbb{R}^r$ such that $z_i = W_rv_i$. We get from (3.3b) and (3.6) that

$$\beta_{i+1}u_{i+1} = AW_rv_i - \alpha_iu_i,$$
$$\alpha_{i+1}v_{i+1} = W_r^{\top}A^{\top}u_{i+1} - \beta_{i+1}v_i,$$

where the second equation comes from $\alpha_{i+1}W_rv_{i+1} = W_rW_r^{\top}A^{\top}u_{i+1} - \beta_{i+1}W_rv_i$. Combining the above two relations with (3.3a), we conclude that the iterative process for generating $u_i$ and $v_i$ is the standard GKB process of $AW_r$ with starting vector $b$ between the two finite dimensional Hilbert spaces $(\mathbb{R}^r, \langle\cdot,\cdot\rangle_2)$ and $(\mathbb{R}^m, \langle\cdot,\cdot\rangle_2)$. Therefore, $\{u_i\}_{i=1}^{k+1}$ and $\{v_i\}_{i=1}^{k+1}$ are two 2-orthonormal bases of the Krylov subspaces

$$\mathcal{K}_{k+1}(AW_r(AW_r)^{\top}, b) = \mathrm{span}\{(AW_rW_r^{\top}A^{\top})^ib\}_{i=0}^{k},$$
$$\mathcal{K}_{k+1}((AW_r)^{\top}AW_r, (AW_r)^{\top}b) = \mathrm{span}\{(W_r^{\top}A^{\top}AW_r)^iW_r^{\top}A^{\top}b\}_{i=0}^{k},$$

respectively; see, e.g., [13, section 10.4]. Then, $W_rW_r^{\top} = C^{\dagger}_{rkhs}$ implies (4.3). Also, $\{z_i\}_{i=1}^{k+1} = \{W_rv_i\}_{i=1}^{k+1}$ is a $C_{rkhs}$-orthonormal basis of $W_r\mathcal{K}_{k+1}((AW_r)^{\top}AW_r, (AW_r)^{\top}b)$ since $W_r$ is $C_{rkhs}$-orthonormal.

Finally, $\{z_i\}_{i=1}^{k+1}$ are $C_{rkhs}$-orthogonal by construction, and by using the relation

$$W_r(W_r^{\top}A^{\top}AW_r)^iW_r^{\top}A^{\top}b = (W_rW_r^{\top}A^{\top}A)^iW_rW_r^{\top}A^{\top}b = (C^{\dagger}_{rkhs}A^{\top}A)^iC^{\dagger}_{rkhs}A^{\top}b,$$

we obtain that $\{z_i\}_{i=1}^{k+1}$ is a $C_{rkhs}$-orthonormal basis of the RKHS-restricted Krylov subspace defined in (4.2). ∎

PROPOSITION 4.5 (gGKB terminate step). *Suppose that the gGKB process terminates at step $k_t = \max_{k\geq 1}\{\alpha_k\beta_k > 0\}$. Let the distinct nonzero eigenvalues of $AC^{\dagger}_{rkhs}A^{\top}$ be $\mu_1 > \cdots > \mu_s > 0$ with multiplicities $m_1, \ldots, m_s$, and the corresponding eigenspaces are $\mathcal{G}_1, \ldots, \mathcal{G}_s$. Then $k_t = q$, where $q$ is the number of nonzero elements in $\{P_{\mathcal{G}_1}b, \ldots, P_{\mathcal{G}_s}b\}$, and*

$$q = \dim(\mathcal{K}_{\infty}(AC^{\dagger}_{rkhs}A^{\top}, b)) = \dim(\mathcal{K}_{\infty}(C^{\dagger}_{rkhs}A^{\top}A, C^{\dagger}_{rkhs}A^{\top}b)), \tag{4.4}$$

*where $K_{\infty}(M, v) = \mathrm{span}\{M^iv\}_{i=0}^{\infty}$ denotes the Krylov subspace of $\{M, v\}$. Moreover, $\sum_{i=1}^{s} m_i = r$ with $r$ being the rank of $A$ and $k_t = q \leq r$.*

*Proof.* First, we prove (4.4) with $q$ being the number of nonzero elements in $\{P_{\mathcal{G}_1}b, \ldots, P_{\mathcal{G}_s}b\}$. Let $g_j = P_{\mathcal{G}_j}b/\|P_{\mathcal{G}_j}b\|_2$ for $1 \leq j \leq s$, and let $g_1, \ldots, g_q \neq \mathbf{0}$ without loss of generality. Note that $\{g_j\}_{j=1}^{q}$ are orthonormal. Let $G_j$ be a matrix with orthonormal columns that span $\mathcal{G}_j$. Note that $P_{\mathcal{G}_j} = G_jG_j^{\top}$. By the eigenvalue decomposition $AC^{\dagger}_{rkhs}A^{\top} = \sum_{j=1}^{s}\mu_jG_jG_j^{\top}$, we have

$$w_i := (AC^{\dagger}_{rkhs}A^{\top})^{i-1}b = \sum_{j=1}^{s}\mu_j^{i-1}G_jG_j^{\top}b = \sum_{j=1}^{q}\mu_j^{i-1}\|P_{\mathcal{G}_j}b\|_2g_j. \tag{4.5}$$

Hence, $\mathrm{rank}\{w_i\}_{i=1}^{\infty} \leq q$. On the other hand, for $1 \leq k \leq q$, setting $\bar{w}_i = \|P_{\mathcal{G}_j}b\|_2g_j$, we have $(w_1 \ldots, w_k) = (\bar{w}_1, \ldots, \bar{w}_q)T_k$ with

$$T_k = \begin{pmatrix} 1 & \mu_1 & \cdots & \mu_1^{k-1} \\ 1 & \mu_2 & \cdots & \mu_2^{k-1} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & \mu_q & \cdots & \mu_q^{k-1} \end{pmatrix} = \begin{pmatrix} T_{k1} \\ T_{k2} \end{pmatrix},$$

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where $T_{k1} \in \mathbb{R}^{k\times k}$ consists of the first $k$ rows of $T_k$, and $T_{k2} \in \mathbb{R}^{k\times(q-k)}$ consists of the remaining rows. Note that $T_{k1}$ is a Vandermonde matrix and it is nonsingular since $\mu_i \neq \mu_j$ for $1 \le i \neq j \le k$. Then, $T_k$ has full column rank, thereby the rank of $\{w_i\}_{i=1}^k$ is $k$ for $1 \le k \le q$, and $\operatorname{rank}\{w_i\}_{i=1}^\infty \ge \operatorname{rank}\{w_i\}_{i=1}^q = q$. Therefore, we have $\dim(\mathcal{K}_\infty(AC_{rkhs}^\dagger A^\top, b)) = \operatorname{rank}\{w_i\}_{i=1}^\infty = q$.

Also, we have $\dim(\mathcal{K}_\infty(C_{rkhs}^\dagger A^\top A, C_{rkhs}^\dagger A^\top b)) = \operatorname{rank}\{C_{rkhs}^\dagger A^\top w_i\}_{i=1}^\infty = q$, where the first equality follows from

$$(C_{rkhs}^\dagger A^\top A)^{i-1} C_{rkhs}^\dagger A^\top b = C_{rkhs}^\dagger A^\top (AC_{rkhs}^\dagger A^\top)^{i-1} b = C_{rkhs}^\dagger A^\top w_i, \tag{4.6}$$

and the second equality follows from $\operatorname{rank}\{C_{rkhs}^\dagger A^\top w_i\}_{i=1}^\infty = \operatorname{rank}\{w_i\}_{i=1}^\infty = q$ since $C_{rkhs}^\dagger A^\top$ is nonsingular on $\operatorname{span}\{w_i\}_{i=1}^\infty$, which is a subset of $\operatorname{span}\{\mathcal{G}_l\}_{l=1}^s$ by (4.5). In fact, $C_{rkhs}^\dagger A^\top$ is nonsingular on $\operatorname{span}\{\mathcal{G}_l\}_{l=1}^s$ because $\{\mathcal{G}_l\}$ are eigenspaces of $AC_{rkhs}^\dagger A^\top$ corresponding to the positive eigenvalues.

Furthermore, (4.6) and the nondegeneracy of $C_{rkhs}^\dagger A^\top$ on $\operatorname{span}\{w_i\}_{i=1}^\infty$ imply that

$$\begin{aligned}\dim(\mathcal{K}_q(C_{rkhs}^\dagger A^\top A, C_{rkhs}^\dagger A^\top b)) &= \operatorname{rank}(\{C_{rkhs}^\dagger A^\top A)^{i-1} C_{rkhs}^\dagger A^\top b\}_{i=0}^{q-1} \\ &= \operatorname{rank}(\{C_{rkhs}^\dagger A^\top w_i\}_{i=1}^q = \operatorname{rank}\{w_i\}_{i=1}^q = q.\end{aligned} \tag{4.7}$$

That is, the vectors $\{C_{rkhs}^\dagger A^\top A)^{i-1} C_{rkhs}^\dagger A^\top b\}_{i=0}^{q-1}$ are linearly independent.

Next, we prove that $k_t = q$. Clearly, $k_t \le q$ since by Proposition 4.4 $\{z_i\}_{i=1}^{k_t}$ are orthogonal and they are in $\mathcal{K}_{k_t}(C_{rkhs}^\dagger A^\top A, C_{rkhs}^\dagger A^\top b) \subset \mathcal{K}_\infty(C_{rkhs}^\dagger A^\top A, C_{rkhs}^\dagger A^\top b)$, whose dimension is $q$. On the other hand, we show next that if $k_t < q$, there will be a contradiction; hence, we must have $k_t = q$. In fact, (3.3b) and (3.6) imply that, for each $1 \le i \le k_t$,

$$\begin{aligned}C_{rkhs}^\dagger A^\top A z_i &= \alpha_i C_{rkhs}^\dagger A^\top u_i + \beta_{i+1} C_{rkhs}^\dagger A^\top u_{i+1} \\ &= \alpha_i(\alpha_i z_i + \beta_i z_{i+1}) + \beta_{i+1}(\alpha_{i+1} z_{i+1} + \beta_{i+1} z_i),\end{aligned}$$

which leads to

$$\alpha_{i+1}\beta_{i+1} z_{i+1} = C_{rkhs}^\dagger A^\top A z_i - (\alpha_i^2 + \beta_{i+1}^2) z_i - \alpha_i \beta_i z_{i-1}.$$

Note that $\alpha_1 \beta_1 z_1 = C_{rkhs}^\dagger A^\top \beta_1 u_1 = C_{rkhs}^\dagger A^\top b$. Combining the above two relations and using $\alpha_k \beta_k > 0$ for all $k \le k_t$, we have $z_k \in \operatorname{span}\{(C_{rkhs}^\dagger A^\top A)^i C_{rkhs}^\dagger A^\top b\}_{i=0}^{k-1}$ for all $1 \le k \le k_t$. Hence, recursively applying $z_i = \frac{1}{\alpha_i \beta_i} C_{rkhs}^\dagger A^\top A z_{i-1} - \frac{1}{\alpha_i \beta_i}(\alpha_{i-1}^2 + \beta_i^2) z_{i-1} - \frac{\alpha_{i-1}\beta_{i-1}}{\alpha_i \beta_i} z_{i-2}$ for all $2 \le i \le k_t$, we can write

$$\alpha_{k_t+1}\beta_{k_t+1} z_{k_t+1} = \sum_{i=0}^{k_t} \xi_i (C_{rkhs}^\dagger A^\top A)^i C_{rkhs}^\dagger A^\top b$$

with $\xi_i \in \mathbb{R}$ and in particular, $\xi_{k_t} = 1/\Pi_{i=1}^{k_t} \alpha_i \beta_i \neq 0$. Now $\alpha_{k_t+1}\beta_{k_t+1} = 0$ implies that $\{(C_{rkhs}^\dagger A^\top A)^i C_{rkhs}^\dagger A^\top b\}_{i=0}^{k_t}$ are linearly dependent, contradicting the fact that they are linearly independent as suggested by (4.7). Therefore, we have $q = k_t$.

Last, to prove that $\sum_{i=1}^s m_i = r$, it suffices to show that $\operatorname{rank}(AC_{rkhs}^\dagger A^\top) = r$ since the eigenvalues of $AC_{rkhs}^\dagger A^\top$ are nonnegative. To see that its rank is $r$, following the proof of Proposition 4.4, we write $AC_{rkhs}^\dagger A^\top = AW_r W_r^\top A^\top$. Since $AW_r = AV_r \Lambda_r^{1/2}$, we only need to prove that $AV_r$ has full column rank. Suppose $AV_r y = \mathbf{0}$ with $y \in \mathbb{R}^r$. By Theorem 2.4 it follows that $A^\top AV_r y = \mathbf{0} \Leftrightarrow BV_r \Lambda_r y = \mathbf{0} \Leftrightarrow y = \mathbf{0}$. Thus, $\operatorname{rank}(AC_{rkhs}^\dagger A^\top) = \operatorname{rank}(AW_r) = r$. ∎

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**4.2. Properties of the iterative solutions.** First, we show the uniqueness of the solution at each iteration of the algorithm.

THEOREM 4.6 (uniqunes of solution at each iteration). *For any $k$-dimensional subspace $\mathcal{S}_k \subseteq \mathcal{R}(C_{rkhs})$, there exists a unique solution to* (3.1)*. Furthermore, there exists a unique solution to* $\min_{x\in\mathcal{S}_k} \|Ax-b\|_2$ *if $\mathcal{S}_k$ is an RKHS-restricted Krylov subspace.*

*Proof.* Let $W_k \in \mathbb{R}^{n\times k}$ that has orthonormal columns and spans $\mathcal{S}_k$. For any $x \in \mathcal{S}_k$, there is a unique $y \in \mathbb{R}^k$ such that $x = W_k y$. The solution to (3.1) should be $x_k = W_k y_k$, where $y_k$ is the solution to

$$\min_{y\in\mathcal{Y}_k} \|W_k y\|_{C_{rkhs}}, \quad \mathcal{Y}_k = \left\{ y : y \in \operatorname*{argmin}_{\widetilde{y}\in\mathbb{R}^k} \|AW_k\widetilde{y} - b\|_2 \right\}.$$

By [10, Theorem 2.1], it has a unique solution $y_k$ iff $\mathcal{N}(C_{rkhs}^{1/2}W_k)\bigcap\mathcal{N}(AW_k) = \{\mathbf{0}\}$. Now we prove $\mathcal{N}(C_{rkhs}^{1/2}W_k) = \{\mathbf{0}\}$. To this end, suppose $y \in \mathcal{N}(C_{rkhs}^{1/2}W_k)$ and $x = W_k y$. Then $x \in \mathcal{S}_k\bigcap\mathcal{N}(C_{rkhs}^{1/2})$. Since $\mathcal{N}(C_{rkhs}^{1/2}) = \mathcal{N}(C_{rkhs}) = \mathcal{R}(C_{rkhs})^\perp$, we get $x \in \mathcal{S}_k\bigcap\mathcal{R}(C_{rkhs})^\perp \subseteq \mathcal{R}(C_{rkhs})\bigcap\mathcal{R}(C_{rkhs})^\perp = \{\mathbf{0}\}$. Therefore, $x = W_k y = \mathbf{0}$, which leads to $y = \mathbf{0}$.

Now suppose that $\mathcal{S}_k$ is an RKHS-restricted Krylov subspace. Since $\{z_i\}_{i=1}^k$ is a $C_{rkhs}$-orthonormal basis of $\mathcal{S}_k$, the solutions to $\min_{x\in\mathcal{S}_k} \|Ax-b\|_2$ can be written as $x_* = Z_k y_*$ with $y_* = \operatorname{argmin}_{y\in\mathbb{R}^k} \|AZ_k y - b\|_2 = \operatorname{argmin}_{y\in\mathbb{R}^k} \|B_k y - \beta_1 e_1\|_2$, where we have used the same derivation as for obtaining (3.10). This implies the uniqueness of $y_*$ since $B_k$ has full column rank. Therefore, there exists a unique solution to $\min_{x\in\mathcal{S}_k} \|Ax-b\|_2$. ∎

The next result gives a characterization of the solution at the terminate step of gGKB.

THEOREM 4.7. *If gGKB terminates at step $k_t$ defined in* (4.1)*, the iterative solution $x_{k_t}$ is the unique least squares solution to* $\min_{x\in\mathcal{R}(C_{rkhs})} \|Ax-b\|_2$.

*Proof.* Following the proof of Proposition 4.4, the solution to $\min_{x\in\mathcal{R}(C_{rkhs})} \|Ax-b\|_2$ is $x_\star = W_r y_\star$ with $y_\star = \operatorname{argmin}_y \|AW_r y - b\|_2$. Since $AW_r$ has full column rank, it follows that $y_\star$ is the unique solution to $W_r^\top A^\top(AW_r y - b) = \mathbf{0}$. Note that $\mathcal{R}(W_r) = \mathcal{R}(C_{rkhs})$. Thus, $x_\star$ is the solution to $\min_{x\in\mathcal{R}(C_{rkhs})} \|Ax-b\|_2$ iff $P_{\mathcal{R}(C_{rkhs})}A^\top(Ax_\star - b) = \mathbf{0}$.

Now we only need to prove $P_{\mathcal{R}(C_{rkhs})}A^\top(Ax_q - b) = \mathbf{0}$ since $k_t = q$ by Proposition 4.5. Using the property $P_{\mathcal{R}(C_{rkhs})} = C_{rkhs}C_{rkhs}^\dagger$, we get from (3.8c)

$$P_{\mathcal{R}(C_{rkhs})}A^\top U_{k+1} = C_{rkhs}(Z_k B_k^\top + \alpha_{k+1}z_{k+1}e_{k+1}^\top).$$

Combining the above relation with (3.10), we have

$$\begin{aligned} P_{\mathcal{R}(C_{rkhs})}A^\top(Ax_q - b) &= C_{rkhs}(Z_q B_q^\top + \alpha_{q+1}z_{q+1}e_{q+1}^\top)(B_q y_q - \beta_1 e_1) \\ &= C_{rkhs}[Z_q(B_q^\top B_q y_q - B_q^\top\beta_1 e_1) + \alpha_{q+1}\beta_{q+1}z_{q+1}e_q^\top y_q] \\ &= \alpha_{q+1}\beta_{q+1}C_{rkhs}z_{q+1}e_q^\top y_q = \mathbf{0}, \end{aligned}$$

since $\alpha_{q+1}\beta_{q+1} = 0$ when gGKB terminates. ∎

The following result shows that the iDARR method is essentially a variant of the CG method; see [11, section 7.1] for the basic properties of CG.

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THEOREM 4.8. *Define*

$$T : (\mathcal{R}(C_{rkhs}), \langle\cdot,\cdot\rangle_{C_{rkhs}}) \to (\mathbb{R}^m, \langle\cdot,\cdot\rangle_2), \quad x \mapsto Ax, \tag{4.8}$$

*where $x$ and $Ax$ are vectors under the canonical bases of $\mathbb{R}^n$ and $\mathbb{R}^m$. Then $x_k$ obtained by iDARR is identical to the $k$th iterative solution of CG applied to the normal equation $T^*Tx = T^*b$ with initial solution $\widetilde{x}_0 = \mathbf{0}$.*

*Proof.* Note that CG is a Krylov subspace method, and with initial solution $\widetilde{x}_0 = \mathbf{0}$, the $k$th iterative solution is

$$\widetilde{x}_k = \min_{x \in \mathcal{K}_k(T^*T, T^*b)} \|b - Tx\|_2, \quad \mathcal{K}_k(T^*T, T^*b) = \operatorname{span}\{(T^*T)^i T^*b\}_{i=0}^{k-1}.$$

Now we prove that, under the canonical bases, it holds that $T^*y = C_{rkhs}^{\dagger} A^\top y$ for any $y \in \mathbb{R}^m$. From the definition of the adjoint operator, we have for any $x \in \mathcal{R}(C_{rkhs})$ that

$$\langle Tx, y\rangle_2 = \langle T^*y, x\rangle_{C_{rkhs}} \quad \Leftrightarrow \quad (Ax)^\top y = x^\top C_{rkhs}(T^*y).$$

Since $P_{\mathcal{R}(C_{rkhs})} x = x$, we have

$$(Ax)^\top y = (AP_{\mathcal{R}(C_{rkhs})}x)^\top y = x^\top P_{\mathcal{R}(C_{rkhs})} A^\top y,$$

and consequently, $x^\top (C_{rkhs}(T^*y) - P_{\mathcal{R}(C_{rkhs})} A^\top y) = \mathbf{0}$. Noticing that $C_{rkhs}(T^*y) - P_{\mathcal{R}(C_{rkhs})} A^\top y \in \mathcal{R}(C_{rkhs})$, it follows that

$$C_{rkhs}(T^*y) = P_{\mathcal{R}(C_{rkhs})} A^\top y = C_{rkhs} C_{rkhs}^{\dagger} A^\top y.$$

Multiplying $C_{rkhs}^{\dagger}$ from the left to both sides of the above equality and using $T^*y \in \mathcal{R}(C_{rkhs})$, we immediately obtain $T^*y = C_{rkhs}^{\dagger} A^\top y$.

With the above property, we have

$$(T^*T)^i T^*b = (C_{rkhs}^{\dagger} A^\top A)^i C_{rkhs}^{\dagger} A^\top b.$$

Therefore, we have $\mathcal{K}_k((T^*T), T^*b) = \mathcal{S}_k$, which is the $k$-dimensional RKHS-restricted Krylov subspace. Since $x_k = \min_{x \in \mathcal{S}_k} \|Ax - b\|_2$, we finally obtain $x_k = \widetilde{x}_k$. ∎

Note that CG has a good regularization effect for linear inverse problems when equipped with an appropriate early stopping rule [11, section 7.3], which indicates the good regularization effect of iDARR. Furthermore, compared with CG, iDARR has the following advantages: (1) CG solves the normal equation $T^*Tx = T^*b$ instead of the least squares problem. This approach involves explicitly using $A^\top A$, which squares the condition number and can lead to numerical instability. (2) For large-scale $A$, computing $\|x_k\|_{C_{rkhs}}$ in CG becomes infeasible, making it impractical to apply the L-curve criterion for early stopping. (3) CG is unstable in finite precision arithmetic, as rounding errors can significantly delay and disrupt its convergence [29]. In contrast, this issue can be avoided in iDARR by orthogonalizing the vectors $\{z_i\}$ to ensure their numerical $C_{rkhs}$-orthogonality.

Theorems 4.7 and 4.8 together demonstrate the semiconvergence behavior of iDARR: the iterative solution $x_k$ initially approaches an optimal regularized solution, but subsequently diverges toward the unstable naive solution. Therefore, an appropriate early stopping rule is necessary to obtain a good regularized solution.

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**5. Numerical examples.** We test iDARR in solving the Fredholm integral equation of the first kind and 2D image deblurring applications.[1]

**5.1. The Fredholm equation of the first kind.** We first examine iDARR in solving the discrete Fredholm integral equation of the first kind. The tests cover two distinct types of spectral decay: exponential and polynomial. The latter is well-known to be challenging, often occurring in applications such as image deblurring. Additionally, we investigate two scenarios concerning whether the true solution is inside or outside the FSOI.

**Three norms in iterative and direct methods.** We compare the $l^2$, $L^2$, and DA-RKHS norms in iterative and direct methods. The direct methods are based on matrix decompositions. These regularizers are listed in Table 2.

The iterative methods differ primarily in their regularization norms. For the $l^2$ norm, we use the LSQR method in the IR TOOLS package [12], and we stop the iteration when $\alpha_i$ or $\beta_i$ becomes negligible to maintain stability. For the $L^2$ norm, we use gGKB to construct solution subspaces by replacing $C_{rkhs}$ by the basis matrix $B$; note that this method is equivalent to the LSQR method using $L = \sqrt{B}$ as a preconditioner in the IR TOOLS package. The direct methods are Tikhonov regularizations using the L-curve method [16].

**Numerical settings.** We consider the problem of recovering the input signal $\phi$ in a discretization of the Fredholm integral equation in (2.1) with $s \in [1,5]$ and $t \in [0,5]$. The data are noisy observations $b = (y(t_1), \dots, y(t_m)) \in \mathbb{R}^m$, where $t_j = j\Delta t$ for $0 \le j \le m$ with $\Delta t = 5/m$. The task is to estimate the coefficient $x = (\phi(s_1), \dots, \phi(s_n)) \in \mathbb{R}^n$ in a piecewise-constant function $\phi(s) = \sum_{i=1}^n \phi(s_i)\mathbf{1}_{[s_{i-1},s_i]}(s)$, where $\mathcal{S} := \{s_i\}_{i=1}^n \subset [1,5]$ with $s_i = 1 + i\delta$ and $\delta = 4/n$. We obtain the linear system (1.1) with $A(j,i) = K(t_j, s_i)\delta$ by a Riemann sum approximation of the integral. We set $m = 500$ and take $n = 100$ except when testing the computational time with a sequence of large values for $n$.

The $\mathbb{R}^m$-valued noise $\mathbf{w}$ is Gaussian $N(0, \sigma^2 \Delta t I_m)$. We set the standard deviation of the noise to be $\sigma = \|Ax\| \cdot \text{nsr}$, where nsr is the noise-to-signal ratio, and we test our methods with $\text{nsr} = \{0.0625, 0.125, 0.25, 0.5, 1\}$.

We consider two integral kernels,

$$\text{(a) } K(t,s) := s^{-2}e^{-st}; \quad \text{(b) } K(t,s) := s^{-1}|\sin(st+1)|, \tag{5.1}$$

which lead to exponential and polynomial decaying spectra, respectively, as shown in Figure 1. The first kernel arises from magnetic resonance relaxometry [4] with $\phi$ being the distribution of transverse nuclear relaxation times.

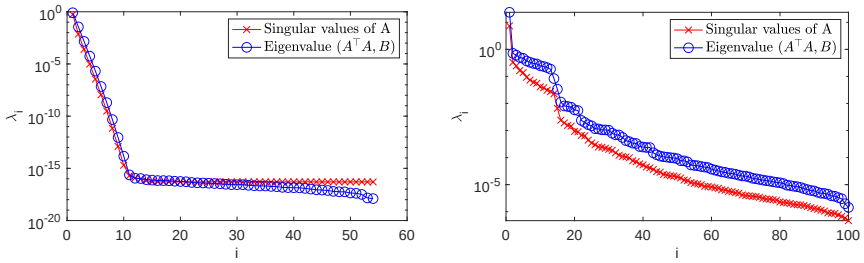
**The DA-RKHS.** By its definition in (2.5), the exploration measure is $\rho(s_i) = \frac{1}{\gamma}\sum_{j=1}^m |K(t_j, s_i)|\delta$ for $i = 1, \dots, n$ with $\gamma$ being the normalizing constant. In other words, it is the normalized column sum of the absolute values of the matrix $A$. The

TABLE 2
*Three regularization norms in iterative and direct methods.*

| Norm | $\|x\|_*^2$ | Iterative | Direct |
|---|---|---|---|
| $l^2$ | $x^\top I_n x$ | IR-l2 | l2 |
| $L^2$ | $x^\top B x$ | IR-L2 | L2 |
| DA-RKHS | $x^\top C_{rkhs} x$ | iDARR | DARTR |

[1]The MATLAB code is available at https://github.com/feilumath/iDARR.

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(a) Exponential decaying spectrum (b) Polynomial decaying spectrum

FIG. 1. *Singular values of A and generalized eigenvalues of* $(A^\top A, B)$ *for kernels in* (5.1).

discrete function space $L^2_\rho(\mathcal{S})$ is equivalent to $\mathbb{R}^n$ with weight $\rho = (\rho(s_1), \ldots, \rho(s_n))$, and its norm is $\langle x, x\rangle_{L^2_\rho} := x^\top \mathrm{diag}(\rho(s_i))x$ for all $x \in \mathbb{R}^n$. The basis matrix for the Cartesian basis of $\mathbb{R}^n$ is $B = \mathrm{diag}(\rho(s_i))$, which is also the basis matrix for step functions in the Riemann sum discretization. The DA-RKHS in this discrete setting is $(\mathcal{N}(A^\top A)^\perp, \langle\cdot,\cdot\rangle_{C_{rkhs}})$.

**Settings for comparisons.** The comparison consists of two scenarios regarding whether the true solution is inside or outside of the FSOI: (i) the true solution is the second eigenvector of $\mathcal{L}_{\overline{G}}$, thus it is inside the FSOI; and (ii) the true solution is $\phi(x) = x^2$, which has significant components outside the FSOI.

For each scenario, we conduct 100 independent simulations, with each simulation comprising five datasets at varying noise levels. The results are presented by a box plot, which illustrates the median, the lower and upper quartiles, any outliers, and the range of values excluding outliers. The key indicator of a regularizer's effectiveness is its ability to produce accurate estimators whose errors decay consistently as the noise level decreases. Since exploring the decay rate in the small noise limit is not the focus of this study, we direct readers to [23, 28] for initial insights into how this rate is influenced by factors such as spectral decay, the smoothness of the true solution, and the choice of regularization strength.

**Results.** We report the results separately according to the spectral decay.

**(i) Exponential decaying spectrum.** Figure 2's top row shows typical estimators of IR-l2, IR-L2, and iDARR and their denoising of the output signal when $\mathrm{nsr} = 0.5$. When the true solution is inside the FSOI, iDARR significantly outperforms the other two in producing a more accurate estimator. However, both IR-L2 and IR-l2 can denoise the data accurately, even though their estimators are largely biased. When the true solution is outside the FSOI, all the regularizers cannot capture the true function accurately, but iDARR and IR-L2 clearly outperform the $l_2$ regularizer. Yet again, all these largely biased estimators can denoise the data accurately. Thus, this inverse problem is severely ill-defined, and one must restrict the inverse to be in the FSOI.

The second row of Figure 2 shows the decay of the residual $\|Ax_k - y\|_2$ as the iteration number increases, as well as the stopping iteration numbers of these regularizers in 100 simulations. The fast decaying residual suggests the need for early stopping, and all three regularizers indeed stop in a few steps. Notably, iDARR consistently stops early at the second iteration for different noise levels, outperforming the other two regularizers in stably detecting the stopping iteration.

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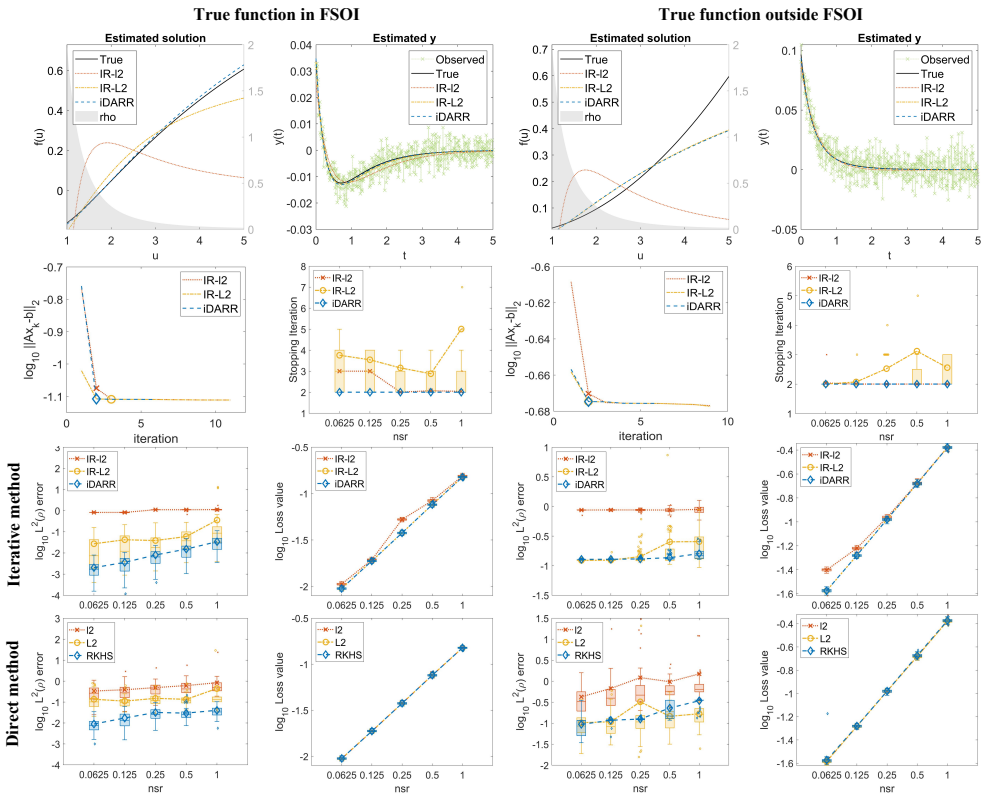



FIG. 2. *Results in the case of exponentially decaying spectrum. Top row: typical estimators of* IR-l2, IR-L2, *and iDARR when* nsr $= 0.5$ *and their denoising of the output signal. Second row: the residual* $\|Ax_k - b\|_2$ *as iteration number* $k$ *increases in one realization when* nsr $= 0.5$, *as well as box plots of the stopping iteration numbers in the* 100 *simulations. Lower two rows: box plots of the estimators'* $L^2_\rho$ *errors and loss function values in the* 100 *simulations.*

The effectiveness of the DA-RKHS regularization becomes particularly evident in the lower two rows of Figure 2, which depict the decaying errors and loss values as the noise-to-signal ratio decreases in the 100 independent simulations. In both iterative and direct methods, the DA-RKHS norm demonstrates superior performance compared to the $l^2$ and $L^2$ norms, consistently delivering more accurate estimators that show a steady decrease in error alongside the noise level. Notably, the values of the corresponding loss functions are similar, underscoring the inherent ill-posedness of the inverse problem. Furthermore, iDARR marginally surpasses the direct method DARTR in producing more precise estimators, particularly when the true solution resides within the FSOI. The performance of iDARR suggests that its early stopping mechanism can reliably determine an optimal regularization level, achieving results that are slightly more refined than those obtained with DARTR using the L-curve method.

**(ii) Polynomial decaying spectrum.** Figure 3 illustrates again the superior performance of iDARR over IR-L2 and IR-l2 in the case of polynomial spectral decay. The second row shows that the slow spectral decay poses a notable challenge to the iterative methods, as the noise level affects their stopping iteration numbers. Also, they all stop early within approximately 12 steps, even though the true solution may lie in a subspace with a higher dimension.

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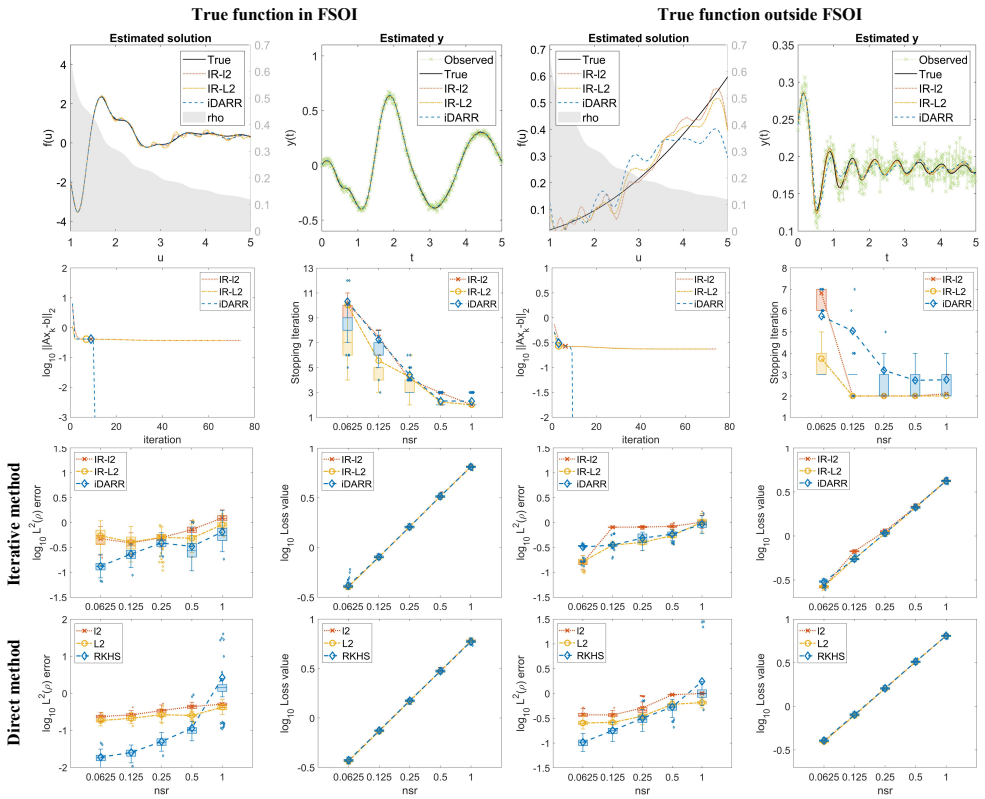



FIG. 3. *Results in the case of polynomial decaying spectrum. Top row: typical estimators of* IR-l2, IR-L2, *and iDARR when* nsr $= 0.0625$ *and their denoising of the output signal. Second row: the residual* $\|Ax_k - b\|_2$ *as iteration number* $k$ *increases in one realization when* nsr $= 0.0625$*, as well as box plots of the stopping iteration numbers the* 100 *simulations. Lower two rows: box plots of the estimators'* $L^2_\rho$ *errors and loss function values in the* 100 *simulations.*

The lower two rows show that iDARR remains effective. It continues to outperform the other two iterative regularizers when the true solution is in the FSOI, and it is marginally surpassed by IR-L2 and IR-l2 when the true solution is outside the FSOI. In both scenarios, the direct method DARTR outperforms all other methods, including iDARR, indicating DARTR is more effective in extracting information from the spectrum with slow decay.

**Computational complexity.** The iterative method iDARR is orders of magnitude faster than the direct method DARTR, especially when $n$ is large. Figure 4 shows their computation time as $n$ increases in 10 independent simulations, and the results align with the complexity order illustrated in subsection 3.3.

In summary, iDARR outperforms IR-L2 and IR-l2 in yielding accurate estimators that consistently decay with the noise level. Its major advantage comes from the DA-RKHS norm that adaptively exploits the information in data and the model.

**5.2. Image deblurring.** We further test iDARR with 2D image deblurring problems, where the task is to reconstruct images from blurred and noisy observations. The mathematical model of this problem can be expressed in the form of the first-kind Fredholm integral equation in (2.1) with $s, t \in \mathbb{R}^2$. The kernel $K(t, s)$ is a function that specifies how the points in the image are distorted, called the point

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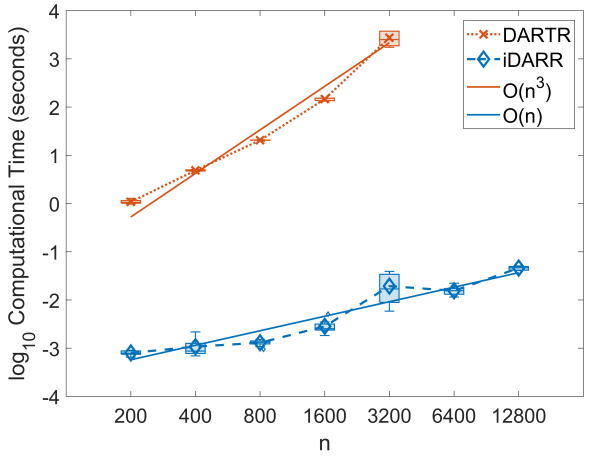
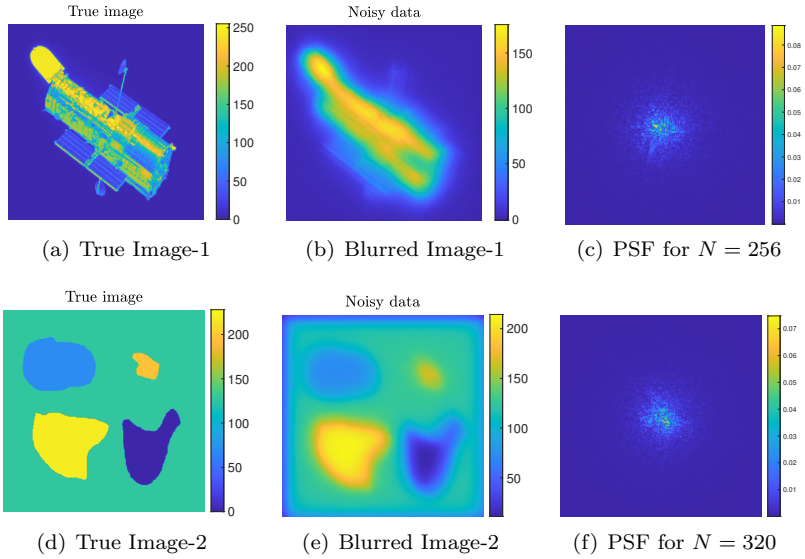



FIG. 4. *Computational time in* 10 *simulations with* $m = 500$.




(a) True Image-1 (b) Blurred Image-1 (c) PSF for $N = 256$

(d) True Image-2 (e) Blurred Image-2 (f) PSF for $N = 320$

FIG. 5. *The true images, noisy images blurred by PRblurspeckle, and the corresponding PSFs.*

spread function (PSF). We choose PRblurspeckle from [12] as the blurring operator, which simulates spatially invariant blurring caused by atmospheric turbulence, and we use zero boundary conditions to construct the matrix $A$. For a true image with $N \times N$ pixels, the matrix $A \in \mathbb{R}^{N^2 \times N^2}$ is a `psfMatrix` object. We consider two images with $256 \times 256$ and $320 \times 320$ pixels, respectively, and set the noise level to be $\text{nsr} = 0.01$ for both images. The true images, their blurred noisy observations, and corresponding PSFs that define matrices $A$ are presented in Figure 5.

We also implement a hybrid variant of iDARR described in (3.16), referred to as hyb-iDARR, and apply it to image deblurring. Figure 6 shows the reconstructed images computed by iDARR, LSQR, hyb-iDARR, and the hyb-l2 methods. Here

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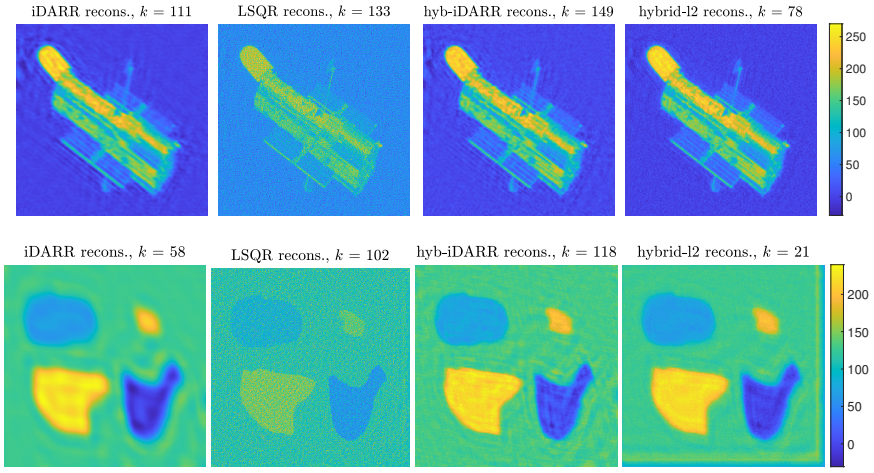
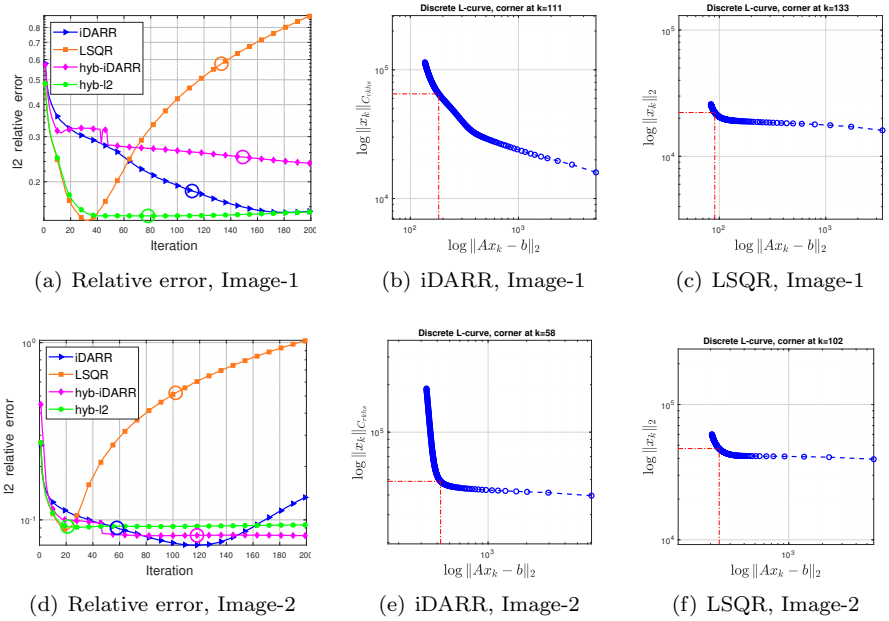



FIG. 6. *The reconstructed images computed by iDARR, LSQR, hyb-iDARR, and hyb-l2 methods.*




(a) Relative error, Image-1 (b) iDARR, Image-1 (c) LSQR, Image-1

(d) Relative error, Image-2 (e) iDARR, Image-2 (f) LSQR, Image-2

FIG. 7. *Relative errors in iteration numbers, where the circles mark the early stopping iterations chosen by the L-curve method presented in the right two columns.*

hyb-l2 applies an $l_2$-norm Tikhonov regularization to the projected problem obtained by LSQR [12]. In the hybrid methods, we terminate the iteration when the difference between two successive solutions becomes very small. The reconstructed solutions are obtained by using the L-curve method for early stopping.

In Figure 7, subfigures (a) and (d) show the relative errors as the iteration number increases, with selection of the early stopping iterations by the L-curve method in the right two columns. Notably, Figure 7 shows that iDARR achieves more accurate reconstructed images than LSQR for both tasks when equipped with an L-curve as the early stopping criterion. The LSQR appears prone to stopping late, resulting

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in lower-quality reconstructions than iDARR. In contrast, iDARR tends to stop earlier than ideal, before achieving the smallest error. The hyb-l2 method exhibits stable convergence for both deblurring tasks. However, for the first task, hyb-iDARR converges slowly, leading to a relatively poor solution even after 200 iterations. We note that a common flaw of hybrid methods is the inability to determine a good $\lambda_k$ at each iteration for certain inverse problems, leading to slow convergence or even nonconvergent solutions; see [34] for further discussion.

The effectiveness of iDARR depends on the alignment of regularities between the convolution kernel and the image, as the DA-RKHS's regularity is tied to the smoothness of the convolution kernel. With the PRblurspeckle featuring a smooth PSF, iDARR obtains a higher accuracy for the smoother Image-2 compared to Image-1. While Image-2 contains piecewise-constant regions with smooth boundaries, the absence of fine-scale texture or high-frequency patterns allows iDARR to better exploit its low-rank, RKHS-constrained subspaces to recover large-scale structures, while simultaneously avoiding overfitting to noise. An avenue for future exploration involves adjusting the DA-RKHS's smoothness to better align with the smoothness of the data.

Additionally, we note that the DA-RKHS regularizer tends to produce softer transitions, which can obscure originally sharp edges. This is inherent to its nonlocal reproducing kernel, as the PRblurspeckle PSF is both smooth and nonlocal. Thus, the DA-RKHS regularizer is not a suitable option for recovering sharp edges from a blurred image. For applications that require edge preservation, a different regularization method or a hybrid metric (e.g., total variation with the DA-RKHS) may be more suitable.

**6. Conclusion.** We have introduced iDARR, a scalable iterative data-adaptive RKHS regularization method, for solving ill-posed linear inverse problems. It searches for solutions in the subspaces where the true signal can be identified and achieves reliable early stopping via the DA-RKHS norm. A core innovation is a generalized Golub–Kahan bidiagonalization procedure that recursively computes orthonormal bases for a sequence of RKHS-restricted Krylov subspaces. We have presented several properties of iDARR, showing its robust regularization performance and the semiconvergence behavior. We have also proposed a hybrid iDARR method that can overcome the semiconvergence of iDARR. Systematic numerical tests on the Fredholm integral equation show that iDARR outperforms the widely used iterative regularizations using the $L^2$ and $l^2$ norms, in the sense that it produces stable accurate solutions consistently converging when the noise level decays. Applications to 2D image deblurring further show iDARR outperforms the benchmark of LSQR with the $l^2$ norm.

**Reproducibility of computational results.** This paper has been awarded the "SIAM Reproducibility Badge: Code and data available" as a recognition that the authors have followed reproducibility principles valued by SISC and the scientific computing community. Code and data that allow readers to reproduce the results in this paper are available at https://github.com/feilumath/iDARR and in the supplementary materials (iDARR-main.zip [local/web 1.28MB]).

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